

FLOW ENGINEERING

**From Value Stream Mapping
to Effective Action**

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FOREWORD BY KAREN MARTIN

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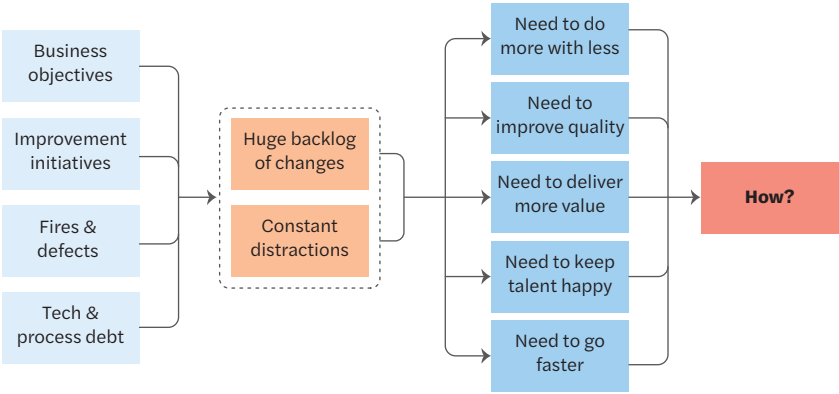
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FIGURE 0.1: Bolt Global Current State Challenge



Every organization grapples with how to address current state challenges in the face of increasing scale.

FIGURE 1.1: The Three Human Costs of Scale

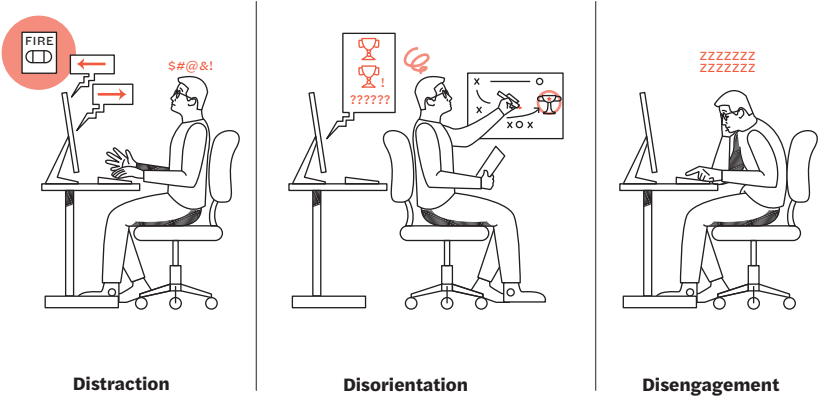
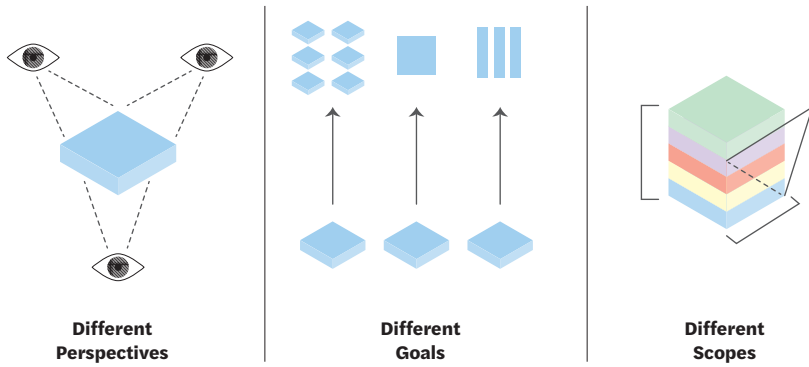
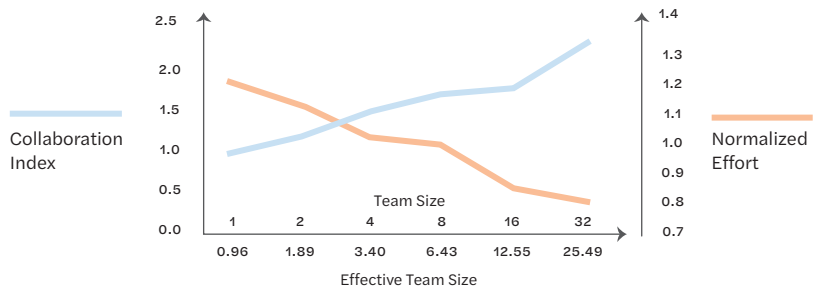


FIGURE 1.2 Misalignment in Teams



Perspectives, goals, and scope vary widely across individuals, across groups, and over time.

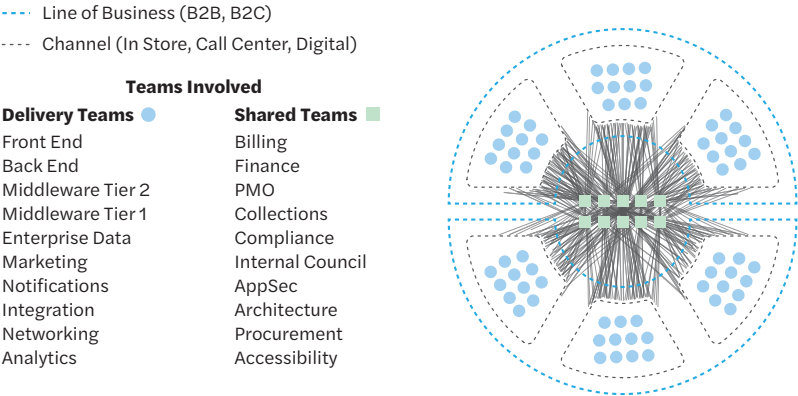
FIGURE 1.3: The Paradox of Scale Illustrated



The graph shows the relative amount of effort by each individual declining and the prevalence of collaboration rising as the size of the team grows (more person-hours).

Adapted from Andrew Mao et al. "An Experimental Study of Team Size and Performance on a Complex Task." *PLoS ONE* 11, no. 4 (April 2016). <https://doi.org/10.1371/journal.pone.0153048>.

FIGURE 1.4 The Checkbox Project: Teams Involved



The effort spanned 10+ delivery teams for each of two lines of business (LOB) with three channels for each LOB, each heavily reliant on shared services.

Source: Kamran Kazempour et al., “The Checkbox Project: Learnings for Organizing for Outcomes,” *The DevOps Enterprise Journal* 5, no. 2 (Fall 2023), <https://itrevolution.com/product/the-checkbox-project/>.

FIGURE 2.1: The Spectrum of Solution Approaches

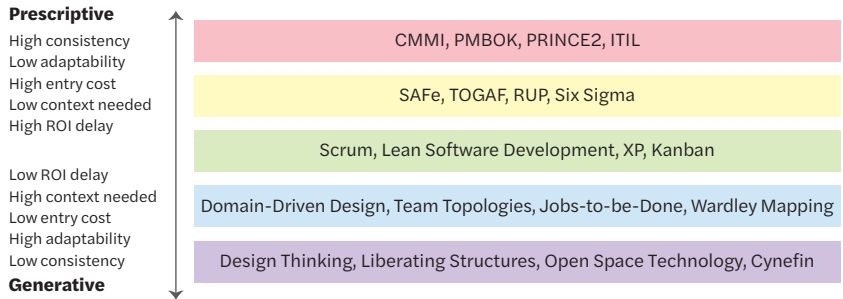


FIGURE 2.2: The Three Gaps to Enabling Large-Scale Collaboration

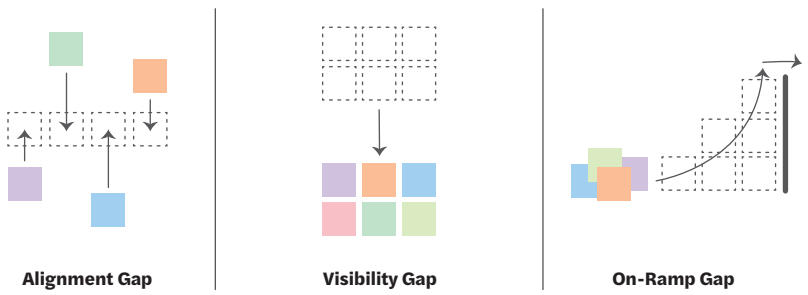


FIGURE 2.3: The Iron Triangle of Constraints

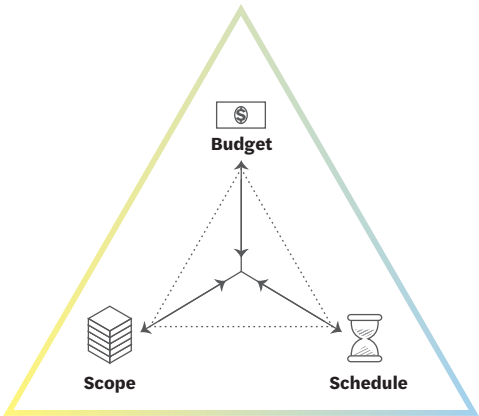
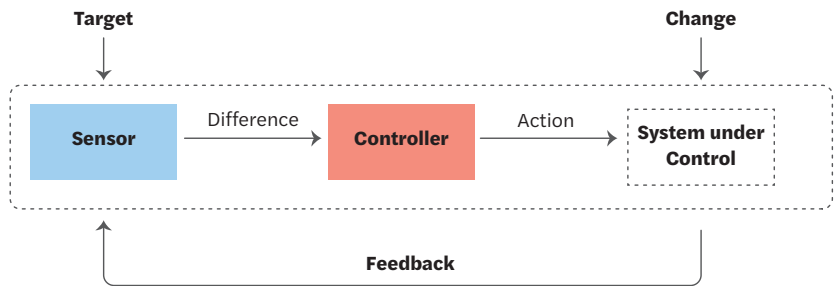
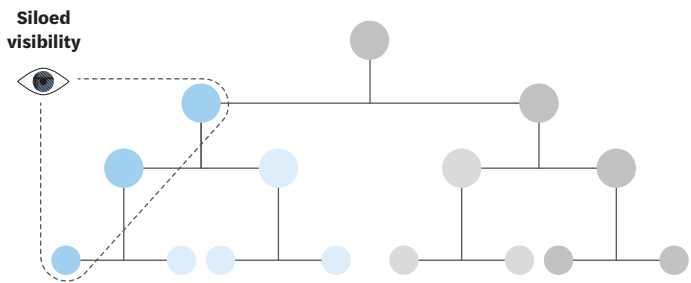


FIGURE 2.4: Cybernetic Feedback Loop



A cybernetic loop represents the flow of navigating change through targeting, sensing, and response.

FIGURE 2.5: A Siloed Organizational Structure



Silos are effective for personnel management but not for cross-organizational flow.

FIGURE 2.6: Effects of Limited Visibility

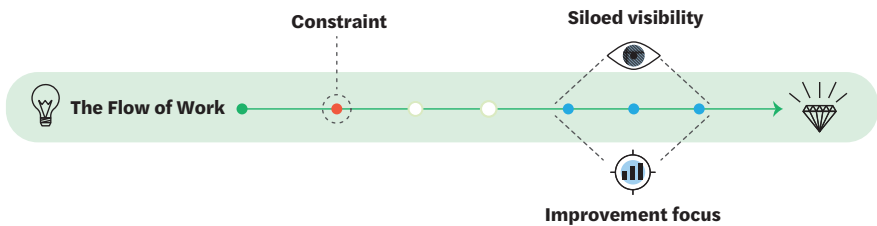
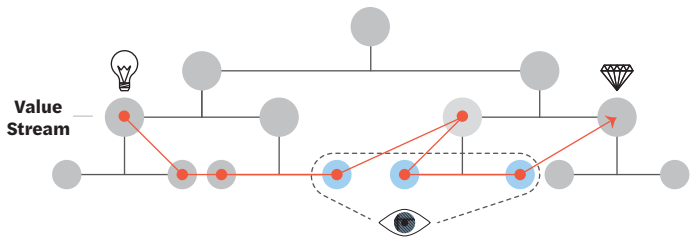


FIGURE 2.7: The Value Stream



The value stream, the full scope of activity needed to satisfy customers, passes through multiple functional groups.

FIGURE 2.8: A Simple Value Stream Map

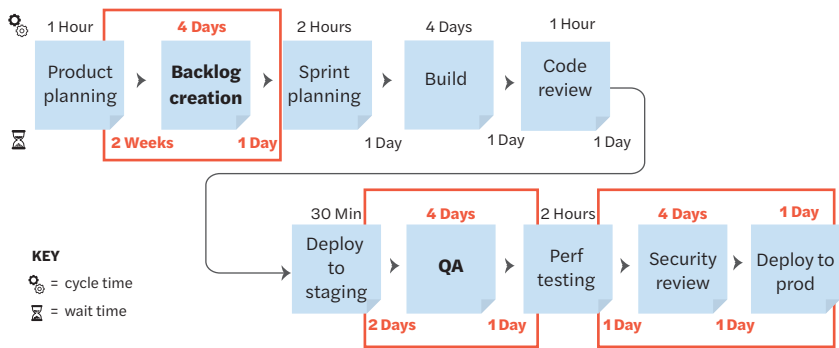
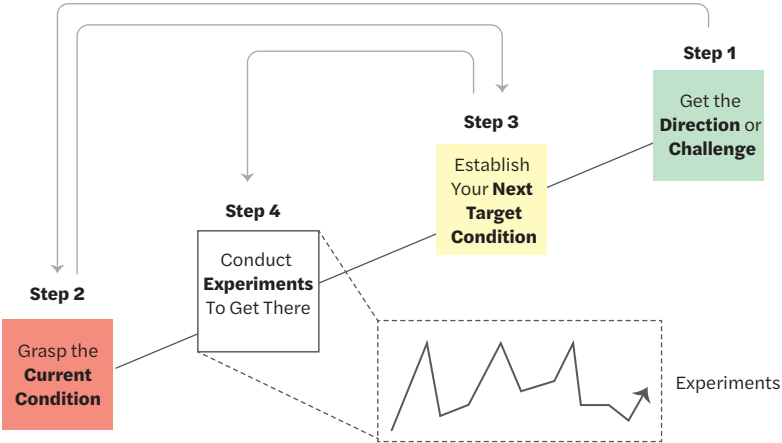


FIGURE 2.9: The Toyota Production System’s Improvement Kata

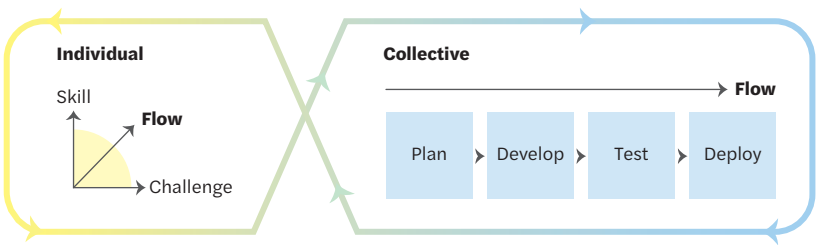


Source: Mike Rother, *Toyota Kata: Managing People for Improvement, Adaptiveness, and Superior Results* (New York: McGraw-Hill, 2009).

TABLE 2.1: Cybernetic Control Systems in Popular Decision Frameworks

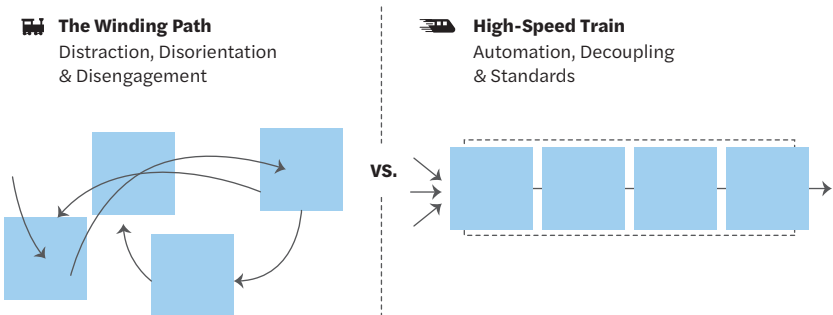
Framework/ Methodology	Target	Sense	Compare	Compute	Act
Lean Startup	Set hypothesis	Measure	Learn	Build	Build
OKRs	Objectives	Monitor key results	Compare to objectives	Compute adjustments	Act on adjustments
V2MOM	Vision	Obstacles	Values	Measures	Methods
Scrum	Sprint goal	Standup	Retrospective	Identified improvements	Improvement
Balanced Scorecard	Strategic objectives	Monitor perspectives	Compare to objectives	Compute strategy	Act on strategy
DMAIC	Define goal	Measure	Analyze	Improve	Control
Lean	Define value	Map the value stream	Measure the value stream against target outcome	Identify constraint & causes	Create flow/ establish pull/strive for perfection
TQM	Quality objectives	Monitor metrics	Compare to objectives	Compute corrective measures	Implement measures
OODA	Mission aim	Observe	Orient	Decide	Act
PDSA/PDCA	Plan	Do/observe	Check/study	Check/study	Act/adapt
Toyota Production System	Operational excellence	Observe (Gemba walks, Andon)	Identify waste (muda, mura, muri)	Kaizen bursts	Implement changes (jidoka, just-in-time, continuous improvement)

FIGURE 3.1: Collective and Individual Flow



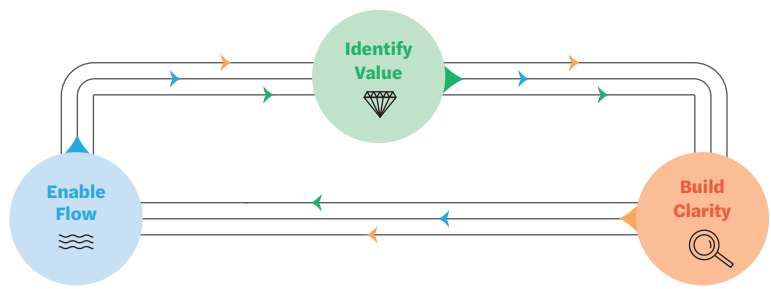
Flow exists across collective activities and within individual work.
The two levels are interdependent.

FIGURE 3.2 Effects of the Three Elements of Action



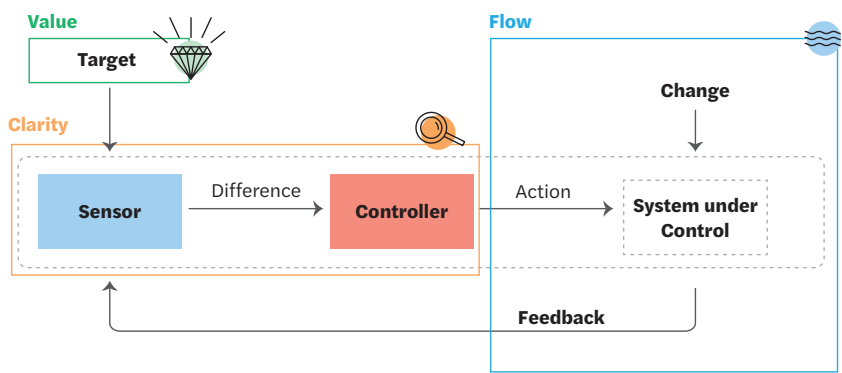
Flow without direction or clarity leads us on a winding path full of waste and confusion. Value and clarity enable you to build high-speed railways for flow.

FIGURE 3.3: The Three Elements of Action



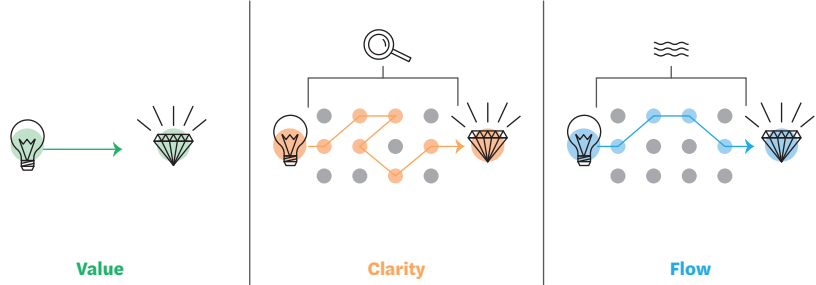
The elements of action—value, clarity, and flow—are interdependent and feed each other.

FIGURE 3.4: Cybernetic Loop with the Elements of Action



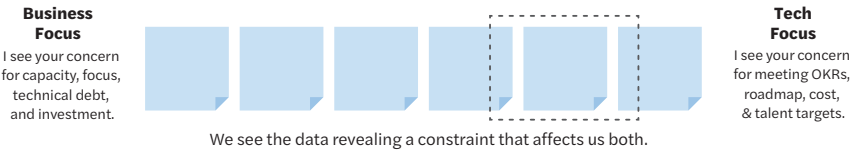
Value, clarity, and flow summarize the cybernetic loop.

FIGURE 3.5: The Elements of Action: Value, Clarity, and Flow



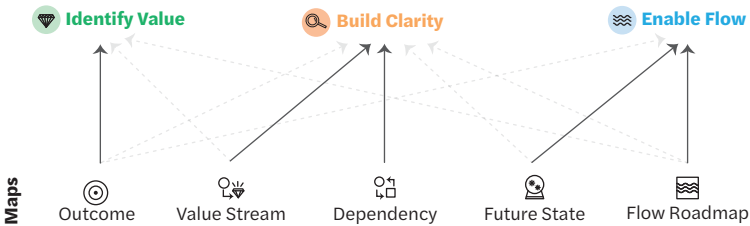
Value is our target; clarity allows us to understand a path to that target; and flow is about optimizing that path.

FIGURE 4.1: The Power of a Map



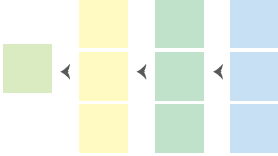
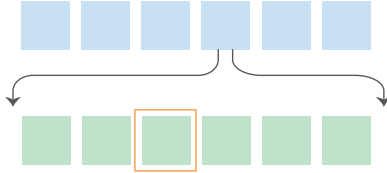
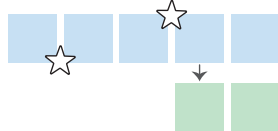
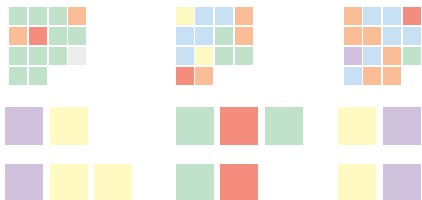
Maps allow individuals to have higher-quality conversations on specific areas of focus.

FIGURE 4.2: Value, Clarity, and Flow Woven through all Flow Engineering Maps



Solid lines indicate a direct contribution, dotted lines indicate indirect contribution.

FIGURE 4.3: Five Maps of Flow Engineering

Maps	Purpose	Simplified Diagram of Maps
 Outcome Map	Discover and align on value	
 Value Stream Map	Find and measure key constraints	
 Dependency Map	Connect constraints to dependencies	
 Future State Map	Design improved flow	
 Flow Roadmap	Create and plot clear next steps	

The full set of Flow Engineering exercises involves creating five maps: an Outcome Map, a Current State Value Stream Map, a Dependency Map, a Future State Value Stream Map, and a Flow Roadmap.

TABLE 4.1: Addressing the Three Gaps through Flow Engineering

Need	Gaps		
	Alignment	On-Ramp	Visibility
Establish and maintain focus on a valuable target	✓	✓	✓
Visual and collaborative in real time	✓	✓	✓
Flexible and easily integrated into current practices	✓	✓	
Accessible to aspiring and novice facilitators		✓	
Inclusive and equitable	✓	✓	✓
Remote friendly	✓	✓	✓
Simple, quick, and affordable	✓	✓	✓
Scalable from pilot to enterprise-wide	✓	✓	✓
Complete program to address flow challenges	✓	✓	✓
Has its own flow (obviously)		✓	

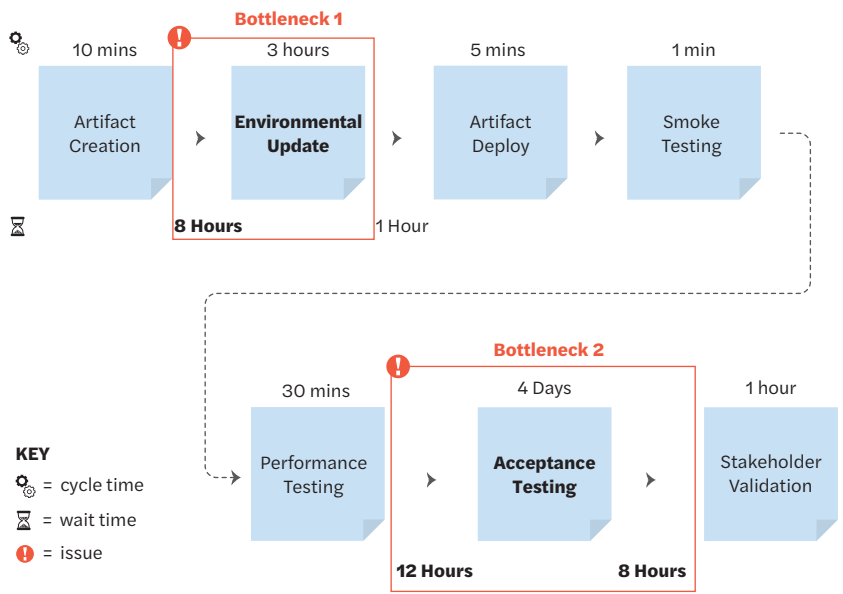
TABLE 4.2: Each Map Serves at Least One of the Elements of Action

Map	Primary goal	Secondary benefits
Outcome Map	Assemble context and identify value	Provide clarity and alignment across multiple perspectives
Current State Value Stream Map	Clearly identify the most impactful constraint	Build clarity on current state workflow and performance
Dependency Map	Build clarity on causal factors	Build an artifact to share with external stakeholders
Future State Value Stream Map	Define improved flow	Identify valuable experiments and actions to improve flow
Flow Roadmap	Prioritize and plan flow improvements	Provide clarity on next steps and flow from today to future goals

TABLE 4.3: General Facilitator Rules of Engagement

✓	Facilitators talk as little as possible outside of guiding the mechanics of the process; they let the participants own most of the dialogue.
✓	Facilitators should caution any individual from talking too long; aim to stay within the duration of an elevator ride. In a larger group, a single sentence constraint can keep the effort moving.
✓	Promote a standard, non-disruptive interruption method (i.e., hand raise) to provide participants a way to voice their thoughts.
✓	Encourage participants to reach out to other participants who haven't spoken to ensure we hear from everyone.
✓	Keep participants comfortable with ranges and guesses; trust but verify with the group.
✓	Facilitators emphasize the need to stay focused on the time horizon in question. If we're looking at the current state, avoid talking about solutions or how things ought to be.

FIGURE 4.4: A Segment of the Full Value Stream Map



This mapping exercise disproved the assumption that automating artifact deployment would make a meaningful improvement. Two other bottlenecks were found instead.

TABLE 4.4: Purposes and Benefits of the Five Flow Engineering Exercises

Mapping Exercise	Purpose	Risk It Averts
Outcome Mapping	Align all members of a team around the value they need to deliver.	Investing in irrelevant improvements.
Current State Value Stream Mapping	Clarify the most likely constraints in an end-to-end workflow.	Optimizing a process that is not the constraint.
Dependency Mapping	Enhance the team’s clarity around likely constraints.	Inadequate understanding of the constraint.
Future State Mapping	Jointly envision a value stream with improved flow.	Failing to create a visible target for change.
Flow Roadmapping	Define the minimum set of steps to achieving improved flow.	Failing to effectively act on insights.

FIGURE 5.1: Outcome Mapping Primarily Serves to Identify Value

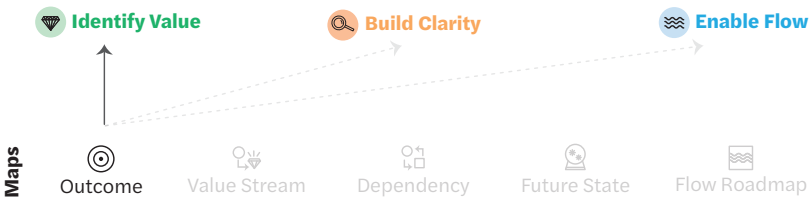


FIGURE 5.2: The Five Stages of Outcome Mapping

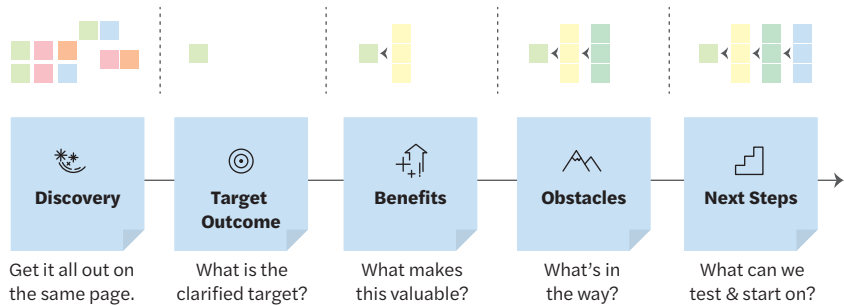
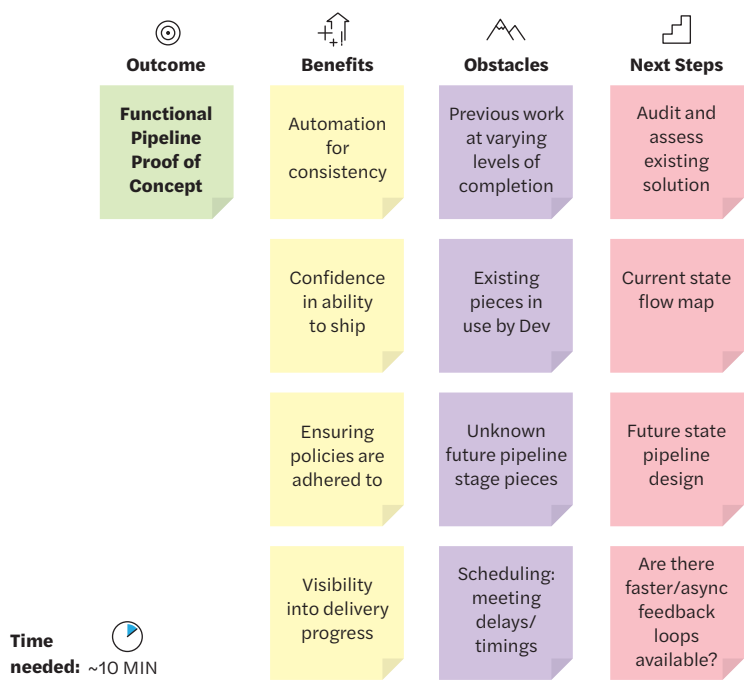


FIGURE 5.3: A Rapid Outcome Map

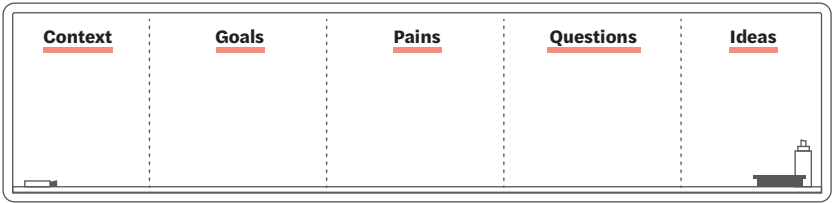


Est. time: Ten minutes to value, clarity, and what's next.

TABLE 5.1: Common Situations That Can Be Transformed into Actionable Outcomes via Outcome Discovery

Issues Inside an Organization	Competitive/External Challenges
Do not have clear OKRs and KPIs.	Financial constraints.
Low organizational performance.	Lacking strategic direction.
eNPS stagnant or dropping.	Lack of clarity around the customer.
Cross-organizational misalignment.	Cannot convince stakeholders to act.
Acting like a “feature factory.”	NPS stagnant or dropping.
Tech debt accumulation.	Customers are leaving or complaining.
Too much WIP.	Customer journey is suboptimal.
Rework or delays.	No data on customer experience.
Improvement initiatives stalled.	Few insights into product usage.
Difficult to prioritize new activities.	Sales are down.

FIGURE 5.4: Five Categories for Outcome Mapping



The five categories or prompts are context, goals, pains, questions, and ideas.

FIGURE 5.5: Stages of Outcome Mapping

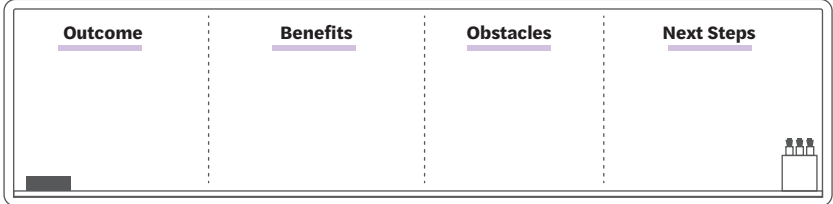
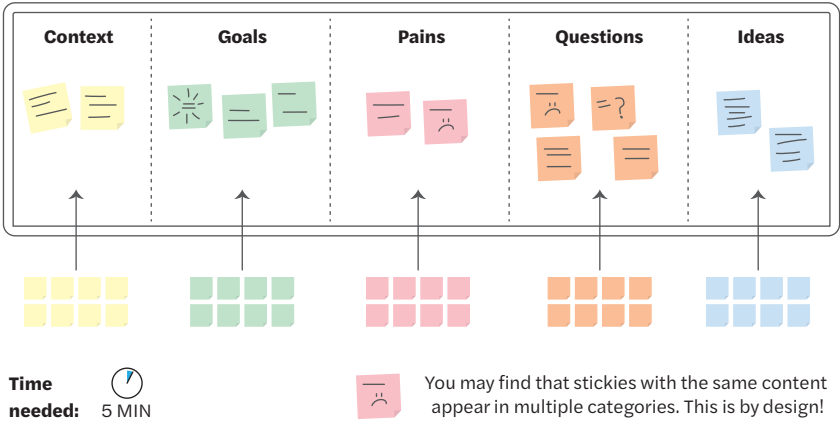
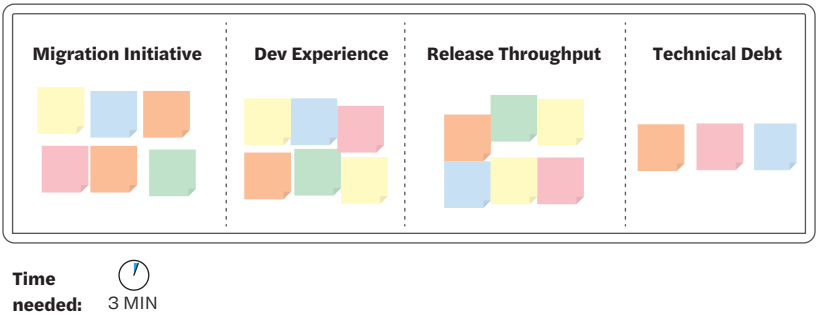


FIGURE 5.6: Example of the Discovery Stage of Outcome Mapping



Discovery stage acts to surface the wide variety of active concerns for a group.

FIGURE 5.7 Notes Organized into Themes



Color coding allows us to group themes while retaining the original item type.

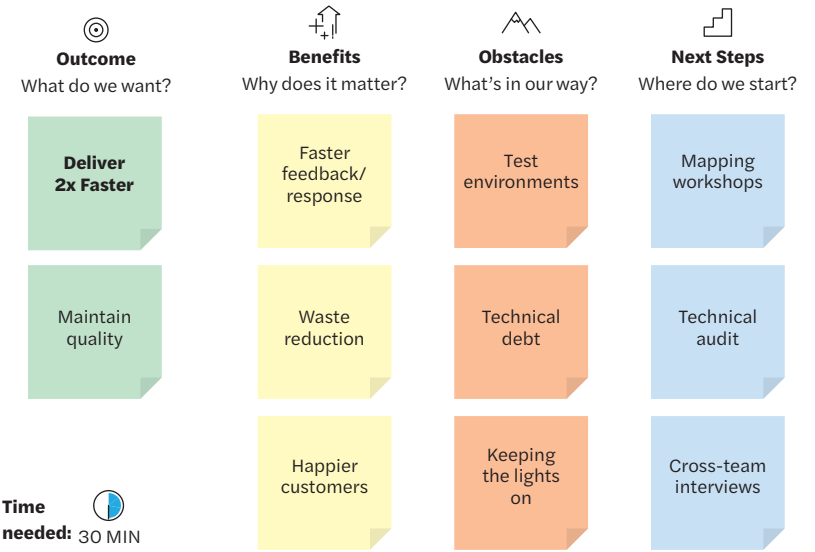
FIGURE 5.8: Example of Voting Using Dots



Time
needed: 2 MIN

Voting allows the group to narrow the focus and align on a single theme to pursue.

FIGURE 5.9: A Completed Outcome Map



Time
needed: 30 MIN

FIGURE 5.10: The Challenge Landscape within Sharon’s Area of Responsibility

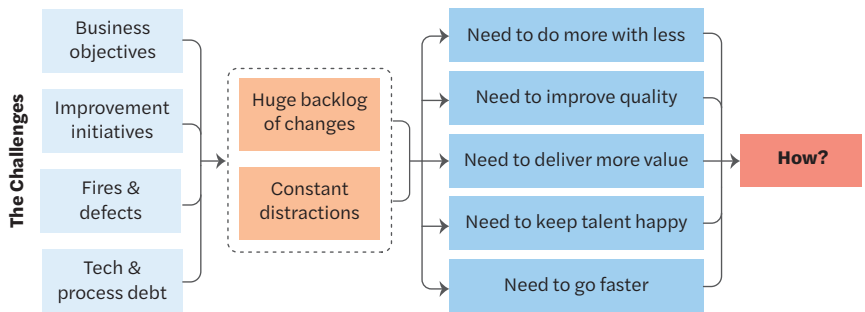


FIGURE 5.11: The Bolt Global Outcome Mapping Process

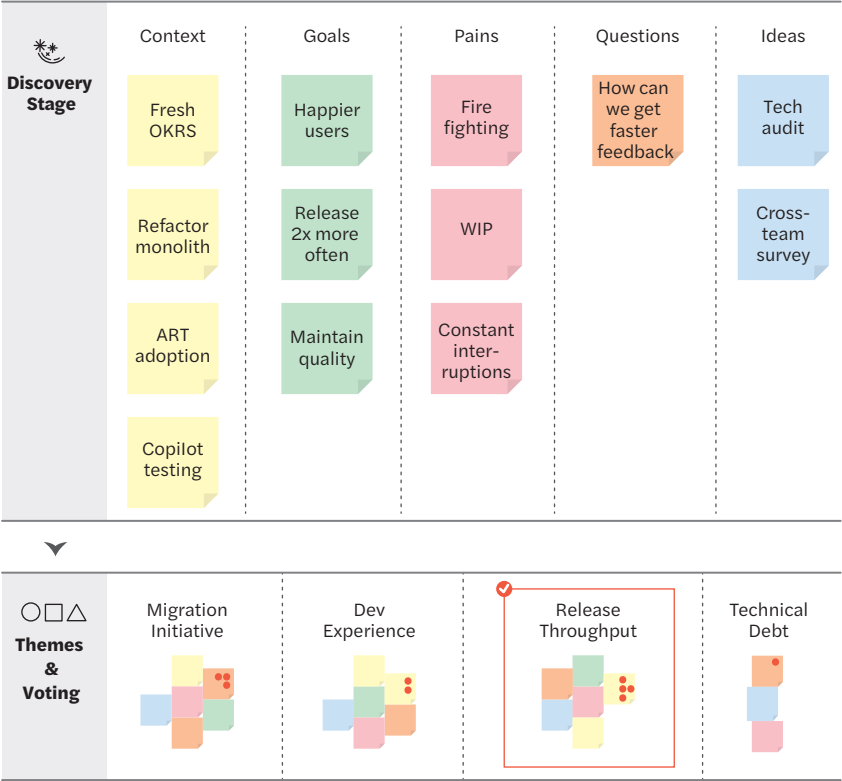


FIGURE 6.1: Value Stream Mapping Primarily Serves to Build Clarity

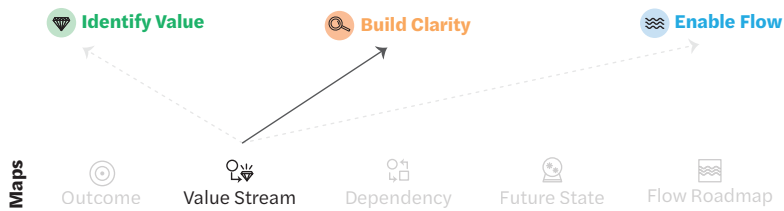


FIGURE 6.2: The Five Stages of Value Stream Mapping

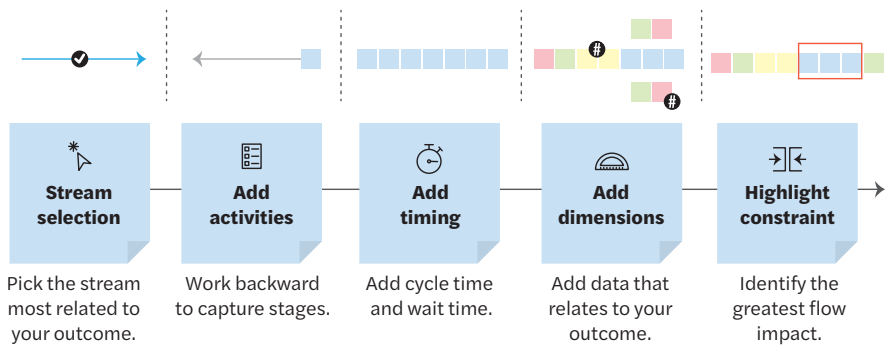
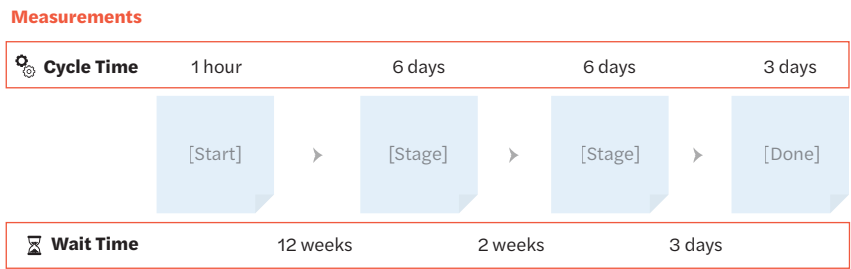
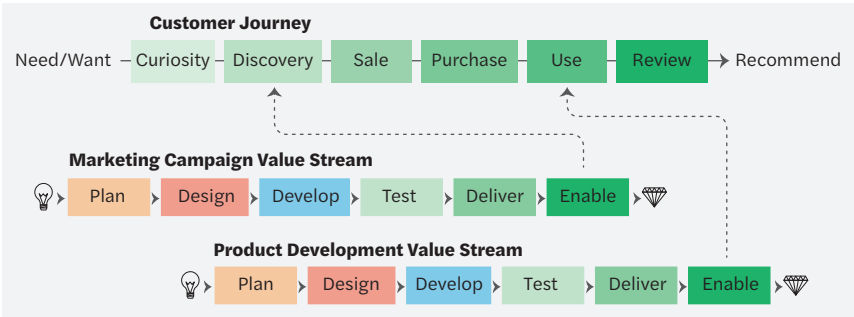


FIGURE 6.3: Simplified Value Stream Map



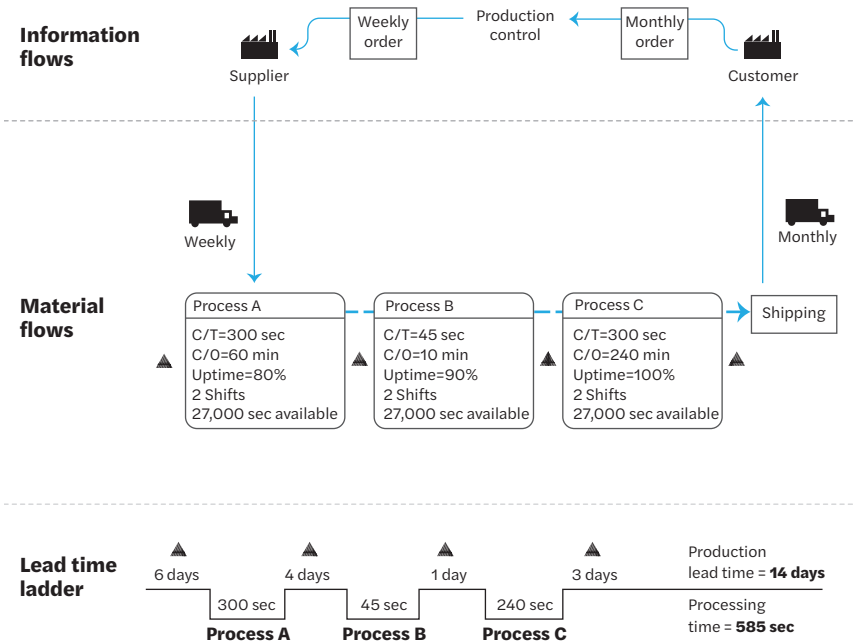
The format of a simplified Value Stream Map, highlighting the key measurements of cycle time and wait time.

FIGURE 6.4: Customer Journey vs. Value Stream



Every part of the customer journey is fed by different value streams in an organization.

FIGURE 6.5: The Traditional Form of Representing a Value Stream Map for Manufacturing



Source: Image adapted from Daniel Penfield, "Value Stream Map Parts,"
Wikimedia Commons, accessed November 2023,
under Creative Commons Attribution-Share Alike 3.0 Unported license.

FIGURE 6.6: A Simple Flow Engineering Value Stream Map

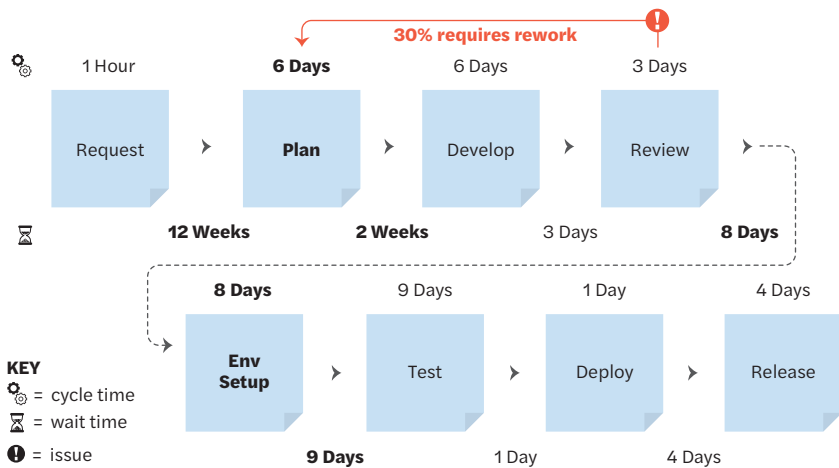
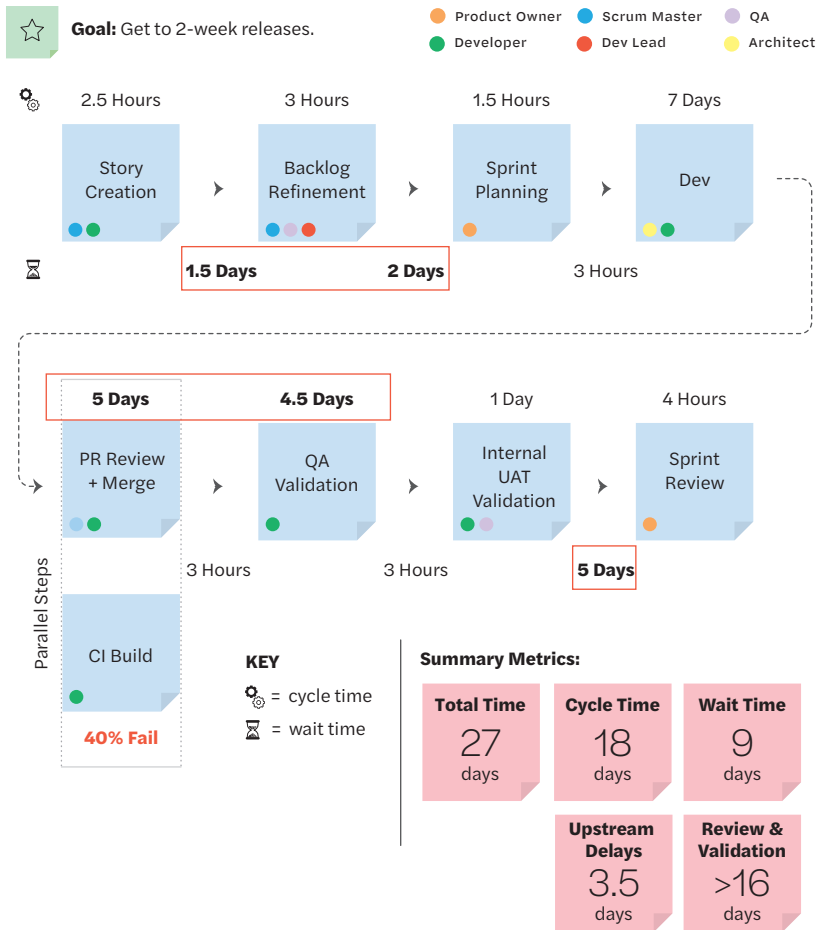


FIGURE 6.7: Example of a Completed Value Stream Map



The value stream map is typically laid out in one horizontal flow. We have broken it here to better fit the page. Parallel steps are represented inside the dotted line box.

FIGURE 6.8: Identifying a Constraint in the Value Stream

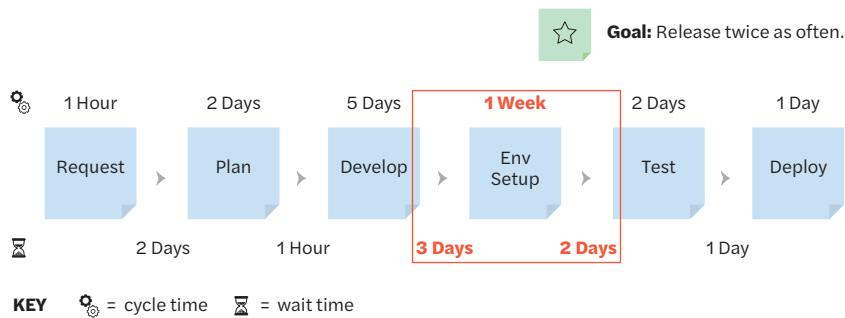


FIGURE 6.9: Example Value Stream Map for Software Development

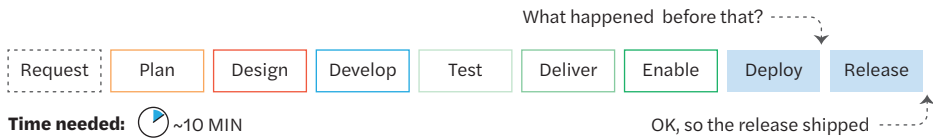
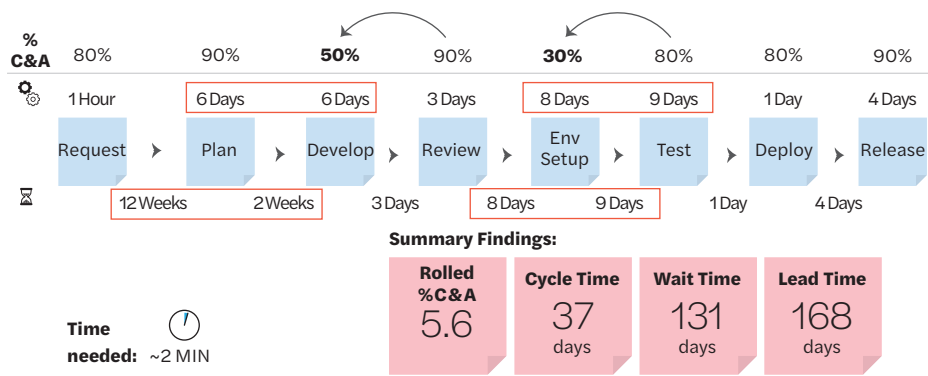


FIGURE 6.10: Sample Value Stream Map Showing Cycle Timing and Wait Timing

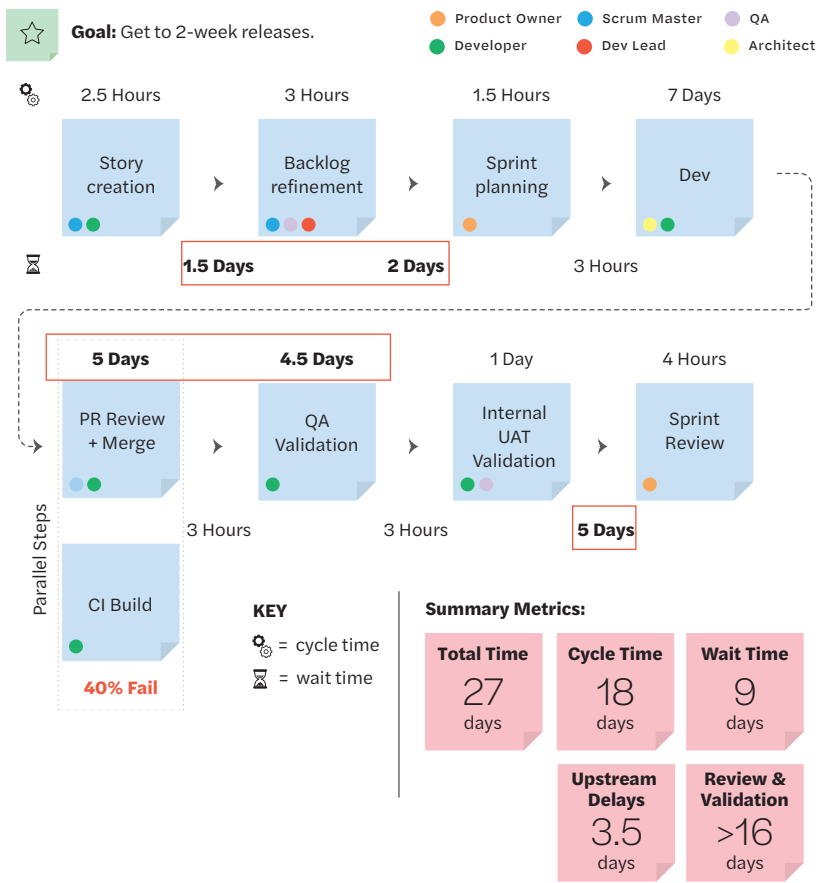


FIGURE 6.11: Annotated Value Stream Map



This annotated version of the map depicts percent complete and accurate (%C&A) for each stage, bold text for particular areas of concern, rework (curved arrows) for stages where errors are detected, and totals for cumulative measures.

FIGURE 6.12: Example Small-Scale Value Stream Map



A small-scale Value Stream Map depicting the target outcome, roles, totals, hot spots, and parallel activities.

FIGURE 6.13: Identifying a Constraint on a Value Stream Map

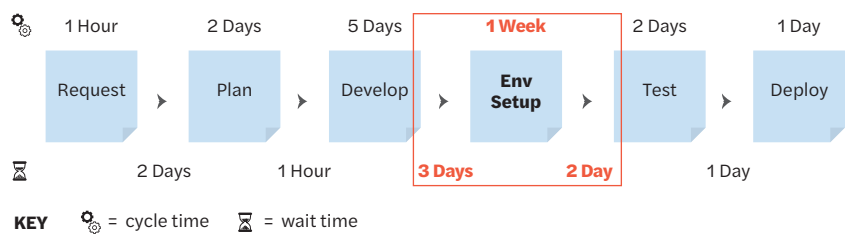


FIGURE 6.14: Value Stream Map with VAT and %C&A

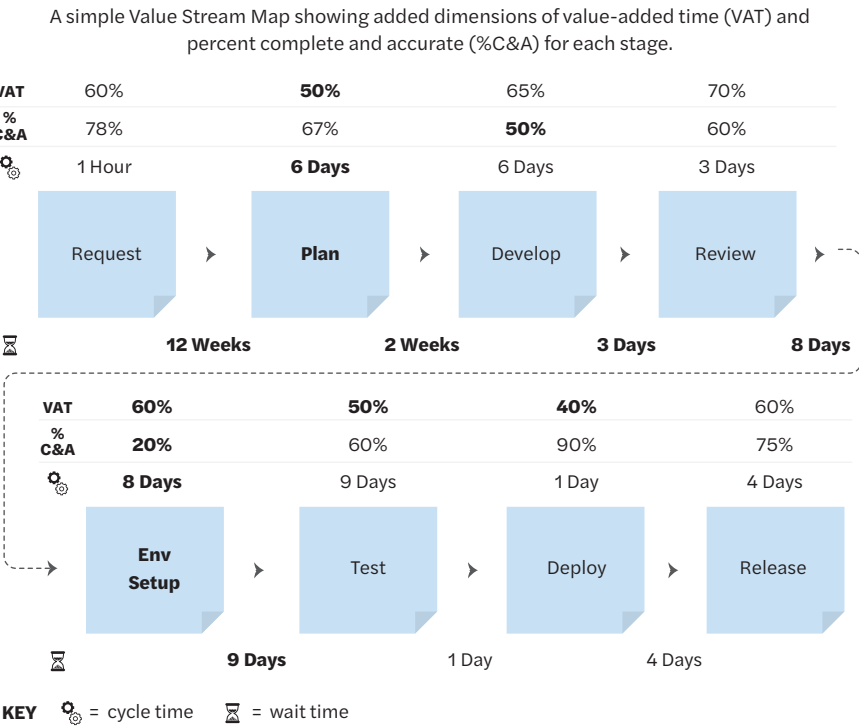
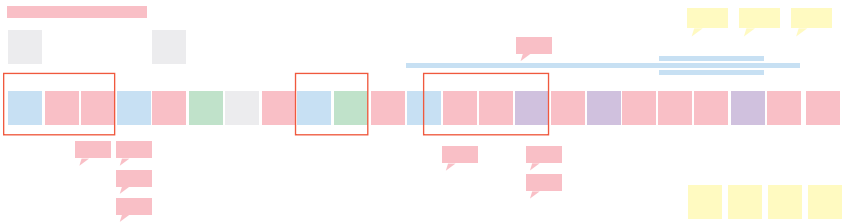


FIGURE 6.15: Sample Value Stream Map with Many Notes



Value Stream Maps can include whatever information is necessary to drive progress and improvement.

Blue	Primarily Value-Adding Activity that creates or enhances value from a customer perspective.	Pink	Primarily Non-Value-Adding Activity doesn't primarily add customer value.
Purple	Testing/Supporting Validating and supporting work—not primarily value-adding.	Green	Coordinating Activity involved in planning, organizing & scheduling.

FIGURE 7.1: Dependency Mapping Primarily Serves to Enhance the Clarity Built by Value Stream Mapping

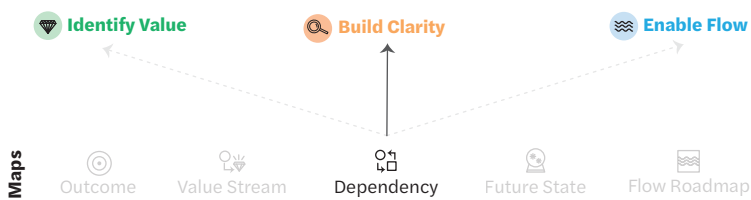


FIGURE 7.2: Dependency Mapping Targets Only Dependencies That Impact the Constraint

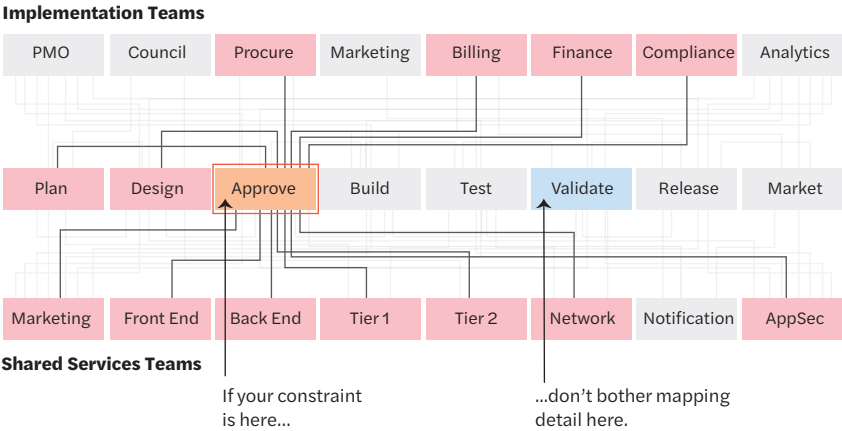
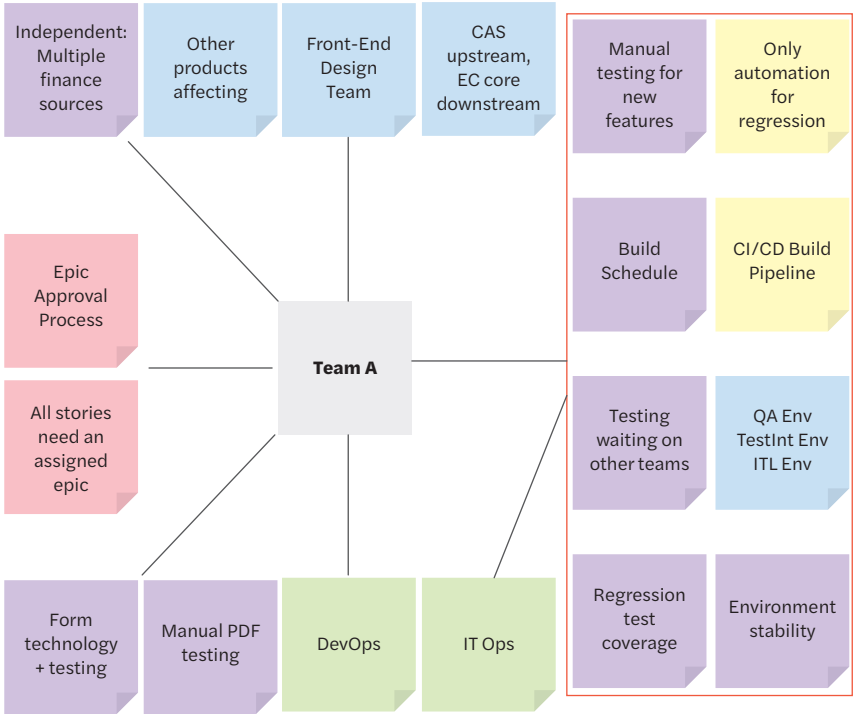
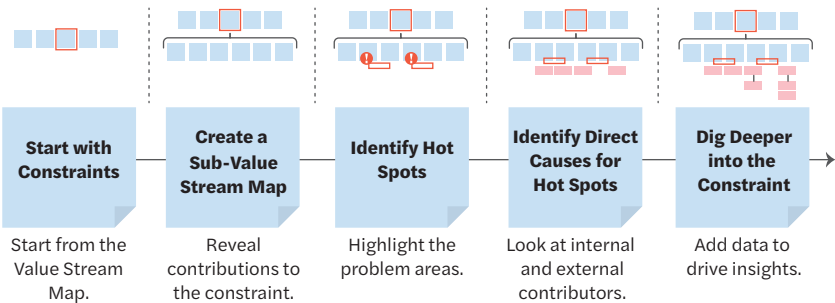


FIGURE 7.3: Example of the Extent of Dependencies



Visualizing all of a team's dependencies can be useful but doesn't identify what might need to improve.

FIGURE 7.4: The Five Stages of Dependency Mapping



KEY ⚙️ = cycle time ⏳ = wait time

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The diagram illustrates the process of shared value, shared clarity, and shared activity for Sharon and Karl. It is structured into three main columns: Target, Perceive & Understand, and Act, leading to the final outcome "Me" or "Us".

- Sharon:**
 - Target:** Represented by a green diamond icon.
 - Perceive & Understand:** Represented by an orange rounded rectangle containing a magnifying glass icon.
 - Act:** Represented by a purple rounded rectangle containing a gear icon.
 - Outcome:** "Me"
- Group:**
 - Shared Value:** Represented by a green diamond icon.
 - Shared Clarity:** Represented by an orange rounded rectangle containing a magnifying glass icon.
 - Shared Activity:** Represented by a purple rounded rectangle containing a gear icon.
 - Outcome:** "Us"
- Karl:**
 - Target:** Represented by a green diamond icon.
 - Perceive & Understand:** Represented by an orange rounded rectangle containing a magnifying glass icon.
 - Act:** Represented by a purple rounded rectangle containing a gear icon.
 - Outcome:** "Me"

Arrows indicate the flow from Target to Perceive & Understand, and from Perceive & Understand to Act. Dashed arrows also point from the final Act stage to the outcome ("Me" or "Us").

Bridging the gap between Sharon and Karl requires sharing value, clarity, and activity.

The diagram illustrates a 3-step process for aligning on value, impact, and constraint. At the top, two circular icons represent a man and a woman, connected by a double-headed arrow, indicating a collaborative relationship. Below them, a horizontal timeline with an arrow at the end shows three steps: 1. Align on **value** (represented by a green circle with a diamond icon), 2. Clarify **impact** (represented by an orange circle with a line graph icon), and 3. Clarify **constraint** (represented by an orange circle with a double-headed arrow icon). A dashed line connects the top icons to the first step, 'Align on value'.

When you're addressing cross-team dependencies, it's helpful to take another pass at the Outcome, Value Stream, and Dependency Maps together with leaders from those teams to align perspectives.

FIGURE 7.8: The Dependency Map Jointly Created by Karl and Sharon

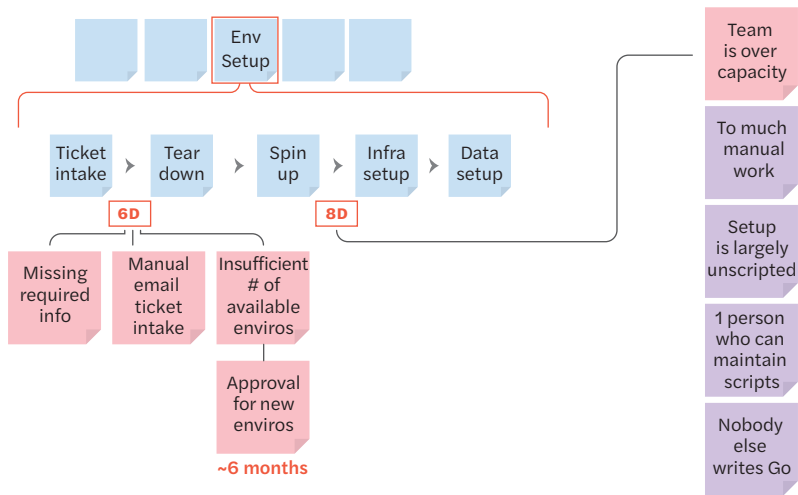


FIGURE 7.9: Suspected Constraint Identified in Value Stream Mapping

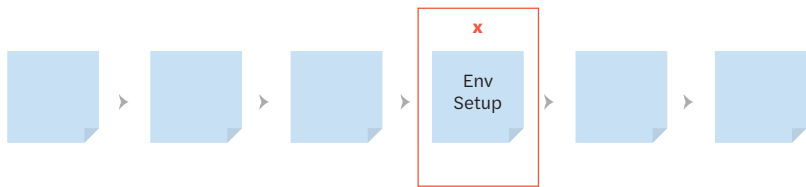
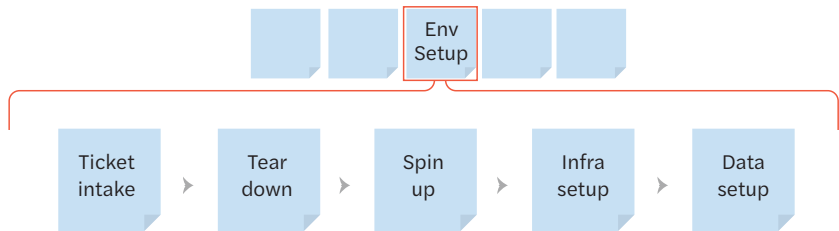


FIGURE 7.10: Example Sub-Value Stream Map



Illustrate the activities that make up the dependency.

FIGURE 7.11: A Sub-Value Stream Map to Illustrate the Dependent Workflow

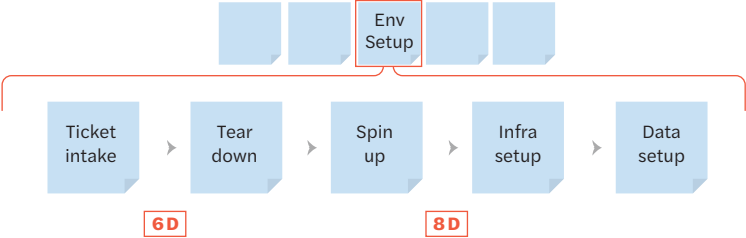
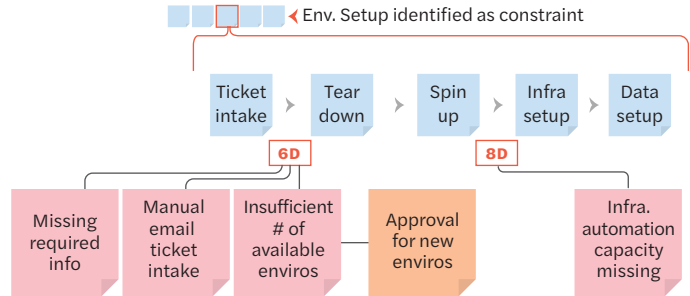


FIGURE 7.12: Direct Causes of Hot Spots



Identify causes of hot spots and areas beyond the group's control. Approval to increase environment capacity is noted in orange as beyond the group's immediate ability to influence.

FIGURE 7.13: Practice the Five Whys to Dig Deeper into the Suspected Constraint

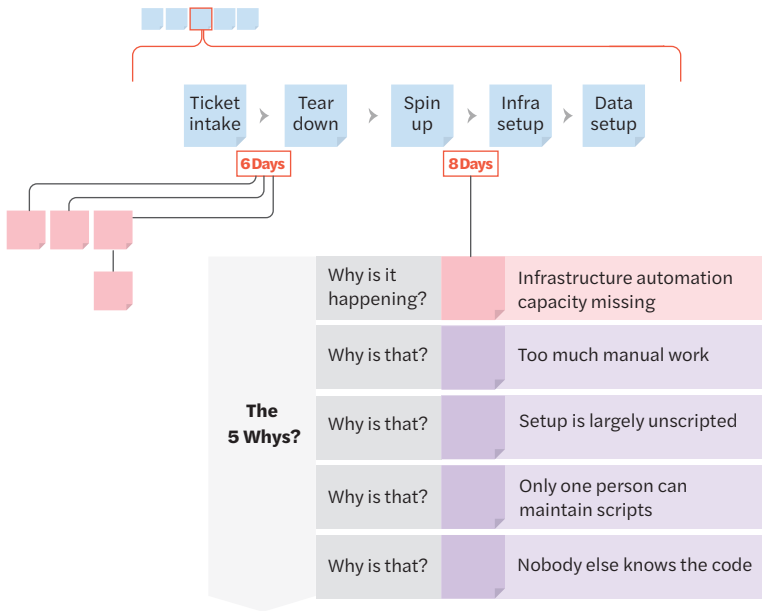


FIGURE 7.14: A Visualization of Dependencies across the Flow of the Checkbox Project

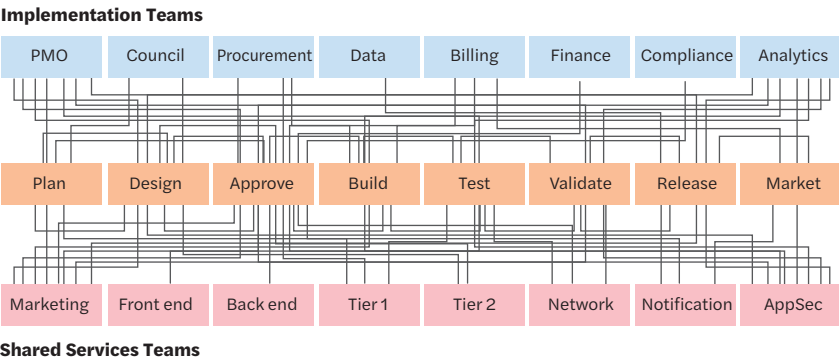


FIGURE 8.1: Future State Mapping Opens the Door to Enabling Flow

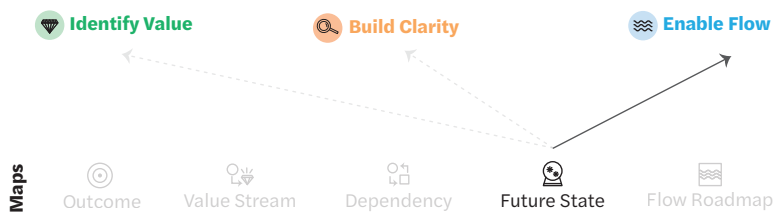
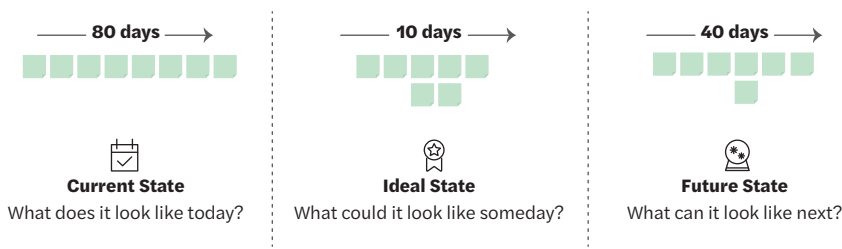


FIGURE 8.2: Current, Ideal, and Future State Maps



Each of these variations of a Value Stream Map represents different potential states for the same workflow.

FIGURE 8.3: Flow Engineering Cycle of Mapping as an Improvement Kata

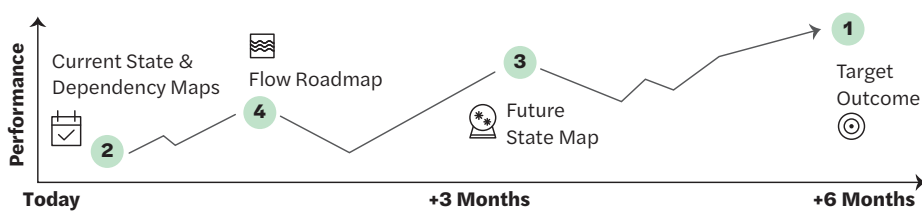


FIGURE 8.4: The Four Stages of Future State Mapping

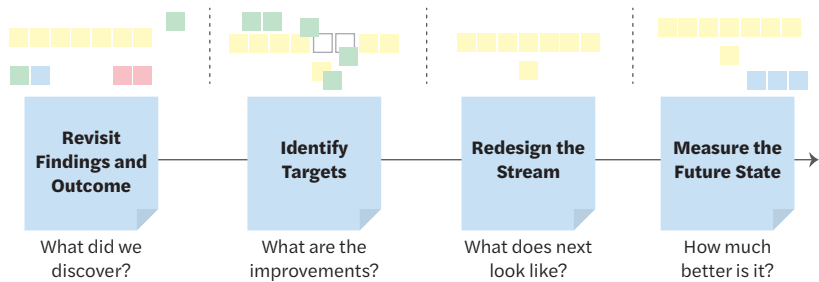


FIGURE 8.5: Current State Map with Bottleneck Highlighted

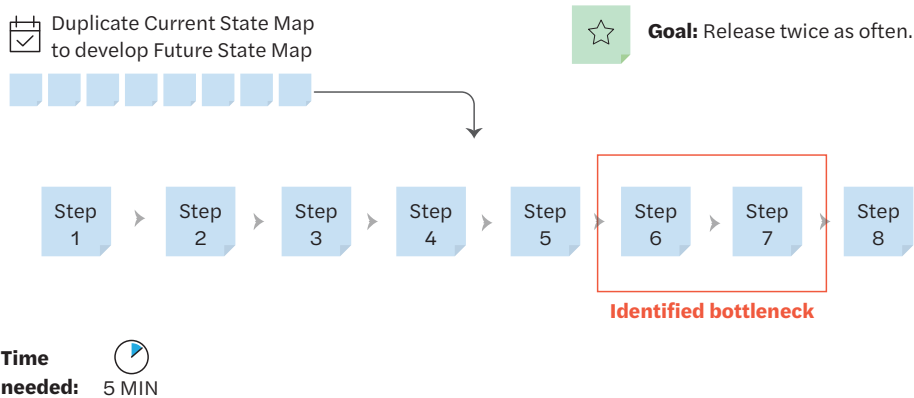


FIGURE 8.6 Future State Map with Improvement Areas Noted in Green

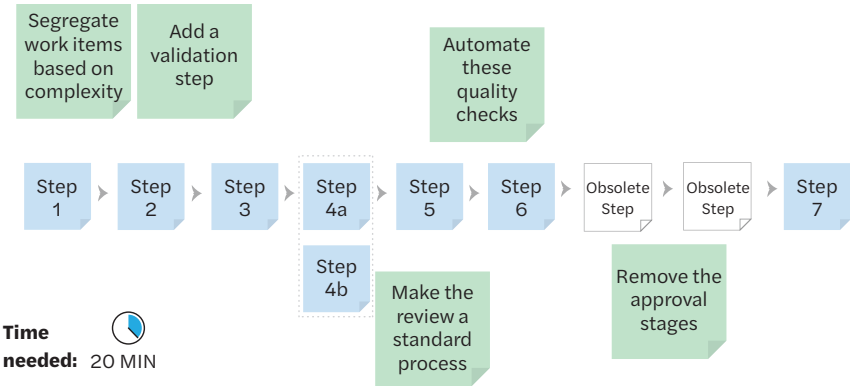


FIGURE 8.7: Future State Map Reduced to Seven Serial Steps and One Parallel Step

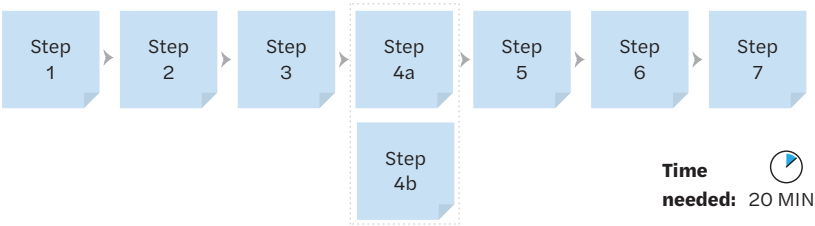


FIGURE 8.8: Future State Map with Cycle Timings

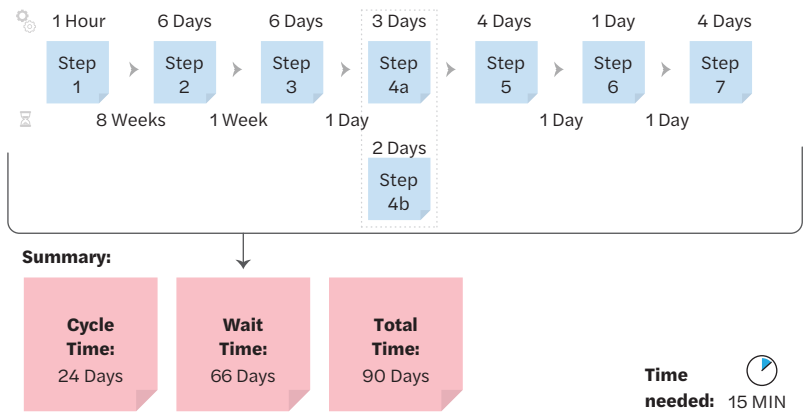


TABLE 8.1: Results of Mapping at a Travel Tech Company

Baseline Flow	Optimized Flow
23 steps	6 steps (including parallel steps)
12 stakeholders	2 stakeholders
12 contributors	4 contributors
8 months	3 months
11 meeting stages	Meetings replaced with decision automation, ownership, collaboration, and coworking
9 call/email steps	Automated weekly customer touch points; relationship calls vs. status calls
9 embedded subprocesses	3 embedded subprocesses

FIGURE 9.1: The Flow Roadmap Primarily Serves to Enable Flow

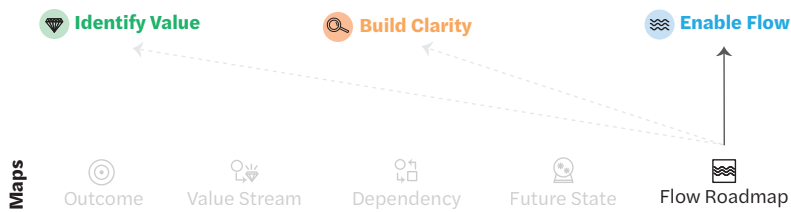


FIGURE 9.2: The Three Stages of a Flow Roadmap

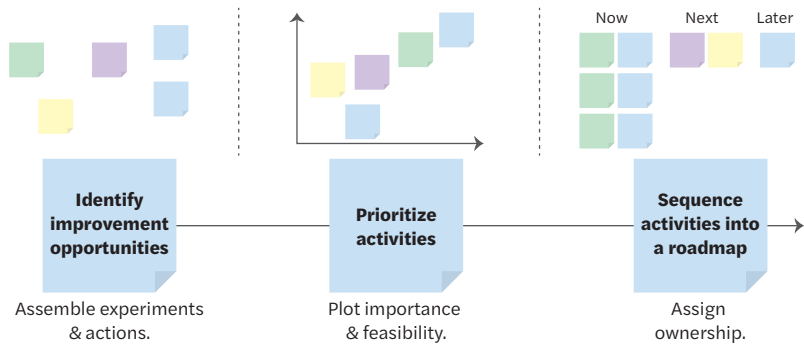
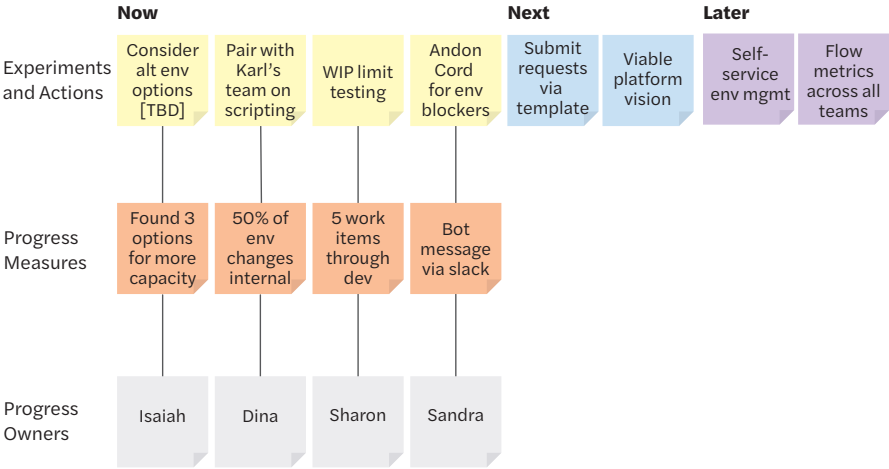


TABLE 9.1: Issues Transformed into Actions/Experiments to Improve Flow at Bolt Global

Issue		Action/Experiment to Improve Performance
Often missing required information.	→	Template for new requests (automate in the future).
Manual email ticket intake.	→	Submit requests via forms into the task management system.
Insufficient number of available environments.	→	Consider alternate options. [TBD]
Team over capacity and constrained.	→	Sharon's team has a contributor who can augment Karl's team capacity.

FIGURE 9.3: Sharon's Completed Flow Roadmap



The completed Flow Roadmap shows actions across three time horizons, measures of progress, and ownership of next steps.

FIGURE 9.4: The Flow Roadmap Board

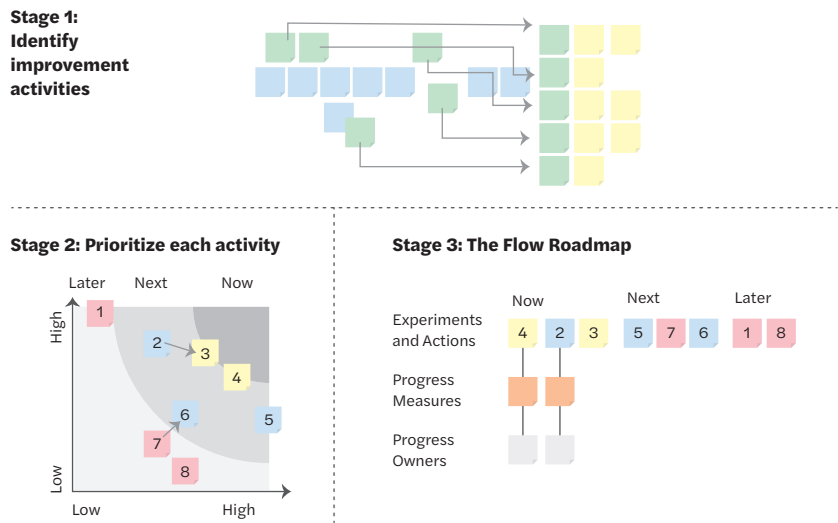
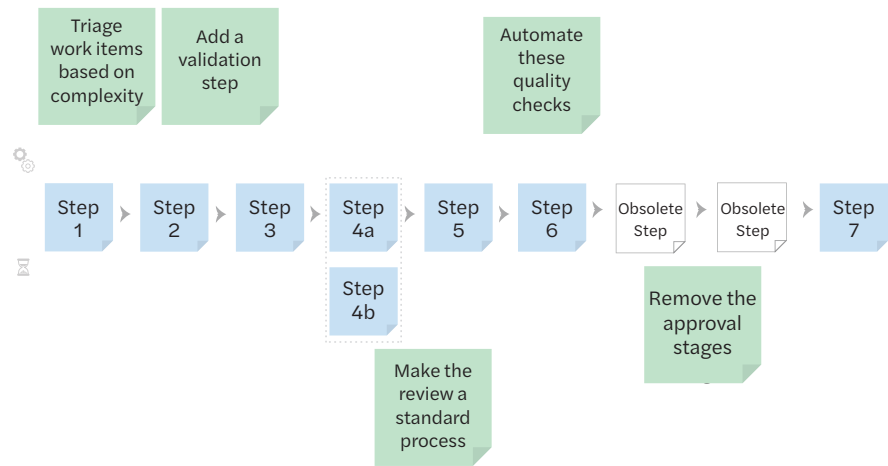


FIGURE 9.5: The Future State Map Showing Improvement Areas



The Future State Map generated in Chapter 8 forms the basis for Flow Roadmapping. The blue sticky notes depict the intended future steps, while the green sticky notes represent areas where change is required to achieve that future state.

FIGURE 9.6: Improvement Opportunities Broken Down into Actions

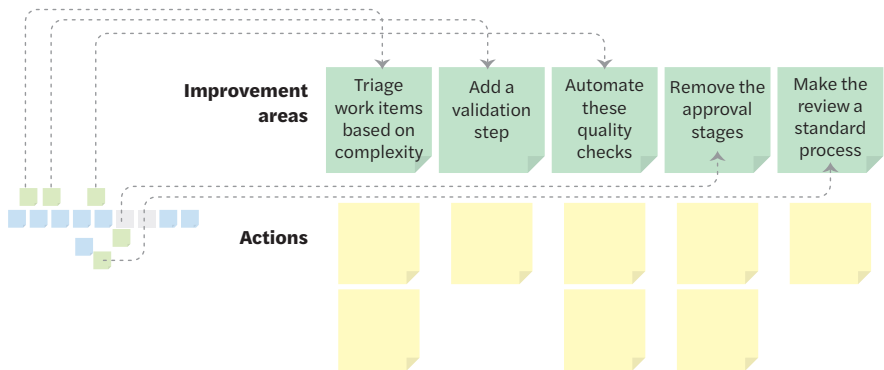


FIGURE 9.7: Possible Improvements Ranked by Importance

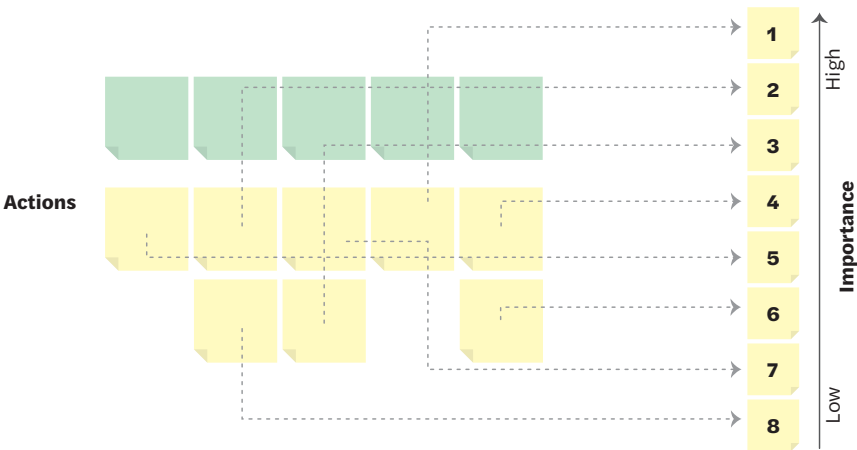


FIGURE 9.8: Plot Feasibility along the Horizontal Axis

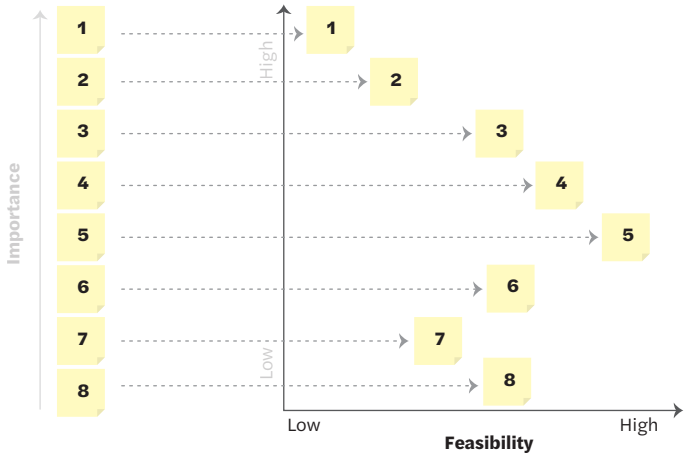


FIGURE 9.9: Indicate Dependencies between Activities

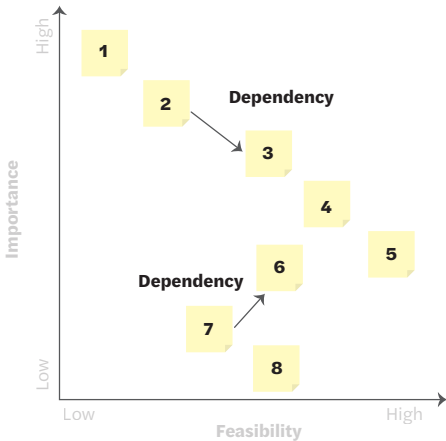
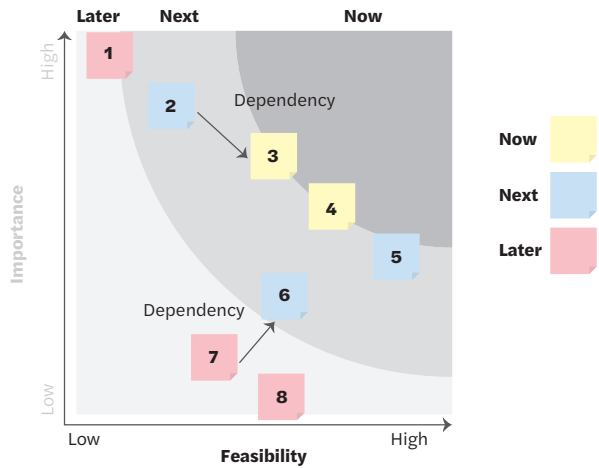


FIGURE 9.10: Improvement Activities Categorized by Time Horizon (Now, Next, Later)



Now is in yellow, next is in blue, and later is in pink.

FIGURE 9.11: Items Arranged in Sequential Order: Now, Next, Later



FIGURE 9.12: Identified Measure of Progress

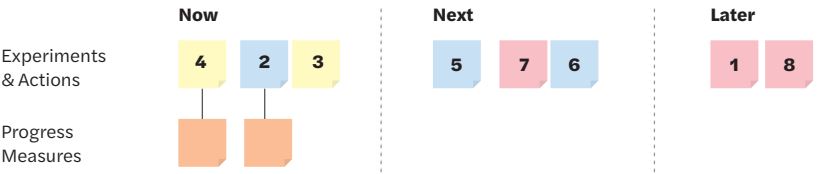


TABLE 9.2: Gartner’s Flow Metrics Cover Technical, Product, and Business Concerns

Technical	Product	Business
Code change size	Lead time	Product cost
Code delivery speed	Cycle time	Product value
Code refactoring rate	Throughput	Return on investment
Code review churn	Work in progress	Product quality
Code quality	Flow efficiency*	Net Promoter Score
Technical debt	Work item profile**	Customer satisfaction

* The percentage of lead time an item is actively being worked on (total cycle time/total lead time).

** Also referred to as Flow Distribution³ and includes features, defects, debt, and risk but can also include experiments or specific subtypes like rework.

Source: Gartner, “Analyze Value Stream Metrics to Optimize DevOps Delivery,” August 2020.

FIGURE 9.13: Example of Adding Assigned Owners

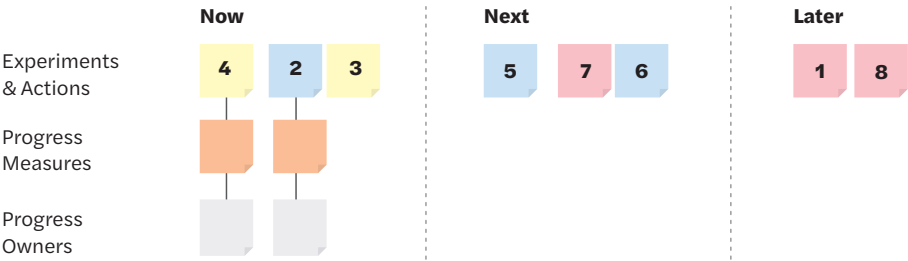


FIGURE 9.14: Flow Roadmap Creation Progressing from Problem Areas and Concerns to Actions

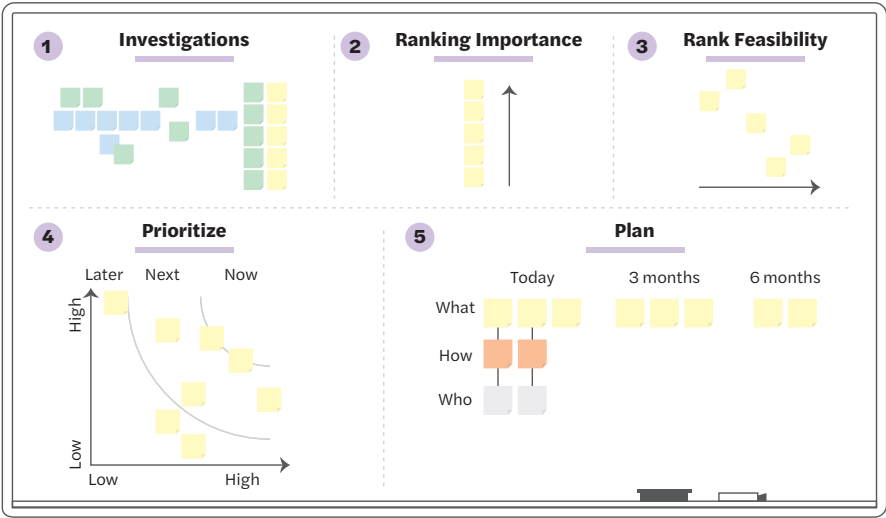


FIGURE 10.1: The Five Principles of Lean

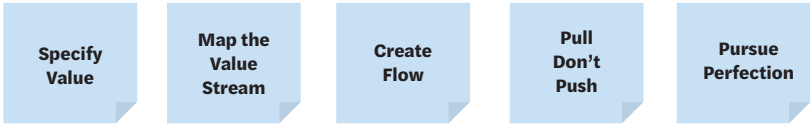


FIGURE 10.2: A Simple Outcome Map Representing Diverse Concerns

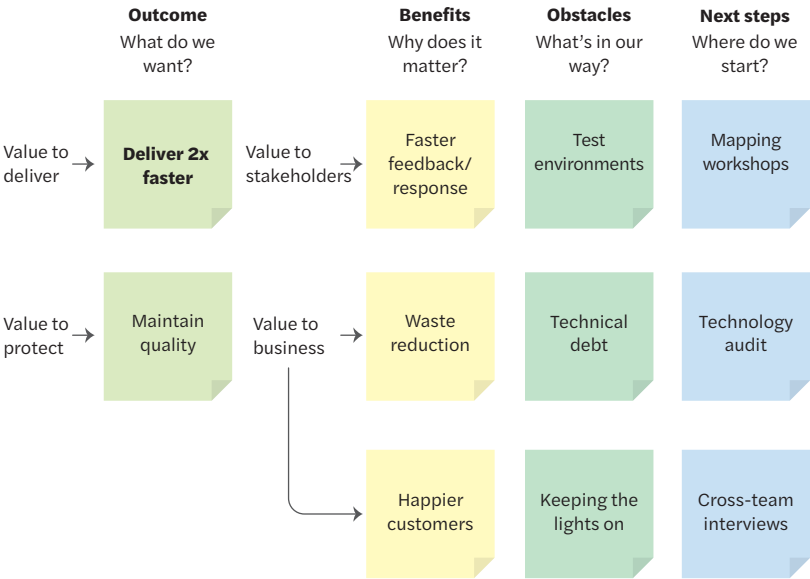


FIGURE 10.3: A Simple Flow Engineering Value Stream Map

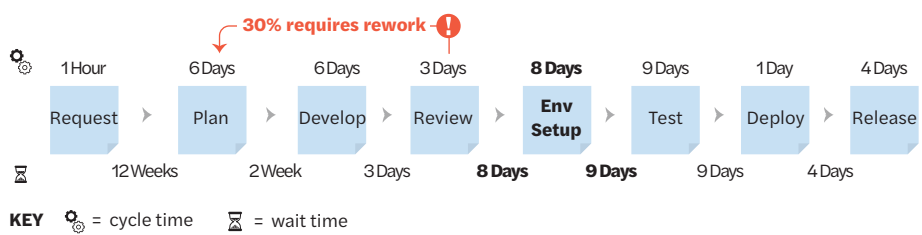


FIGURE 10.4: Value Stream Map Highlighting Wait Times

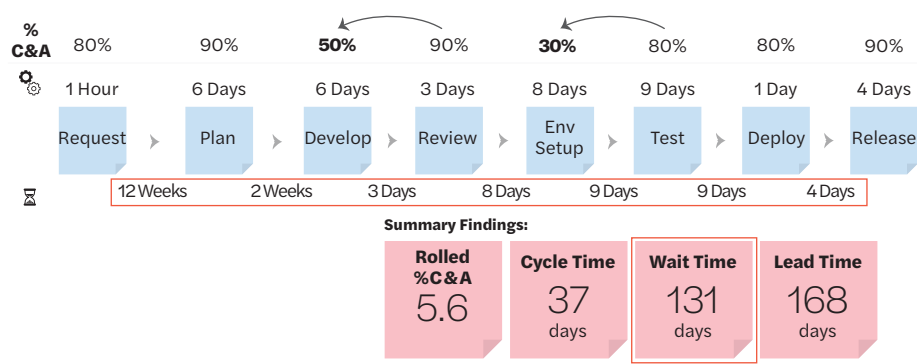
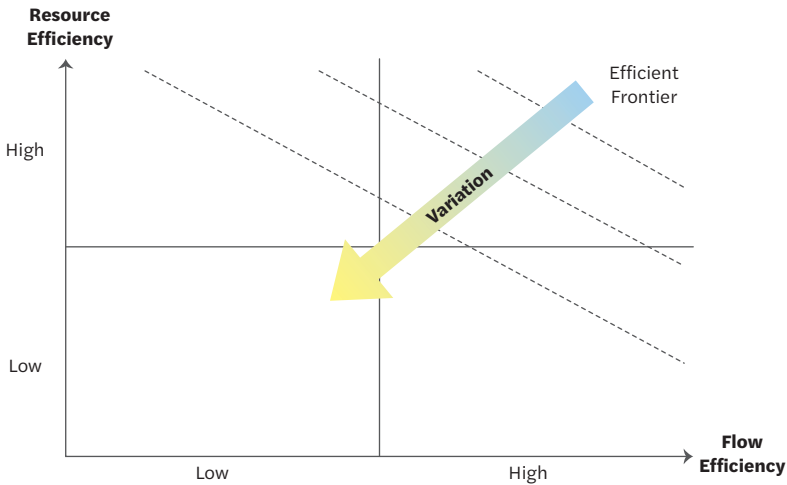


FIGURE 10.5: The Efficiency Matrix



The trade-off between resource efficiency and flow efficiency leads to an Efficient Frontier: a range of possible utilization options, shown as the dotted line in the upper right. The greater the variation in the work, the more that Efficient Frontier is pushed down. *Source: Modig and Åhlström, This is Lean, 105.*

FIGURE 10.6: Theory of Constraints Step 1: Identify the Constraint

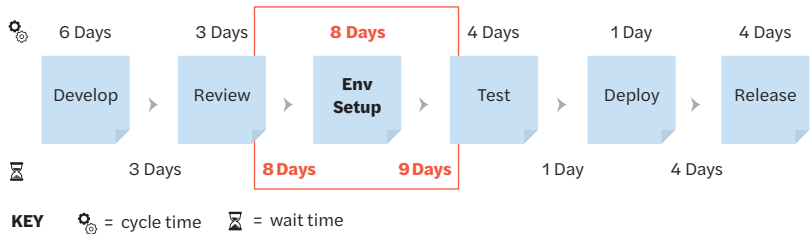


FIGURE 10.7: Theory of Constraints Step 2: Exploit the Constraint

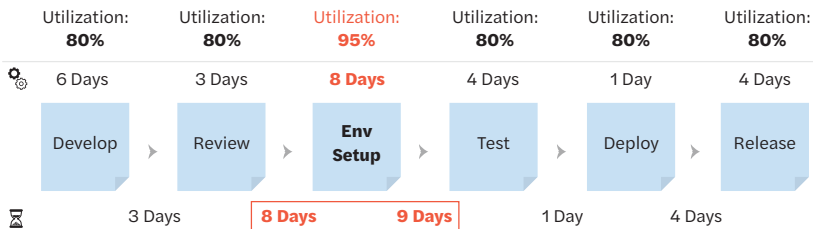


FIGURE 10.8: Theory of Constraints Step 3: Subordinate Everything Else to the Constraint

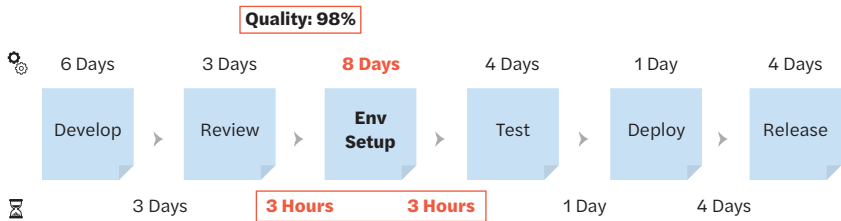


FIGURE 10.9: Theory of Constraints Step 4: Elevate the Constraint

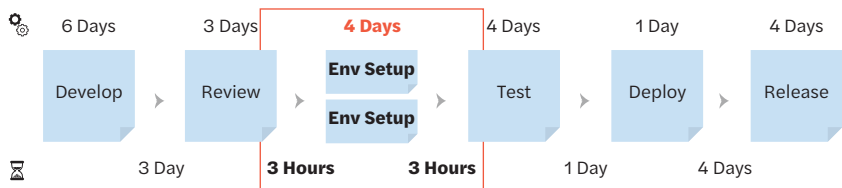


FIGURE 10.10: Excess Work in Progress Is a Sign of Lacking Pull Systems

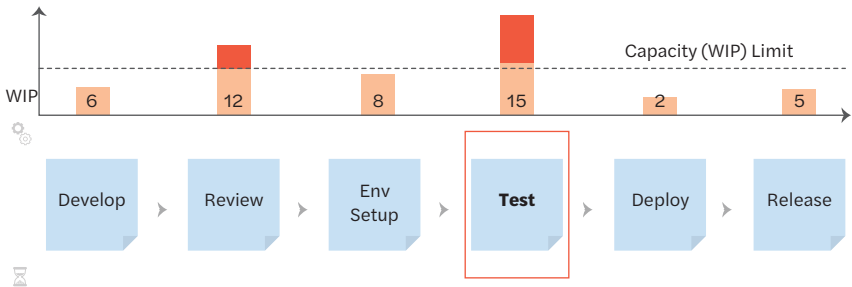
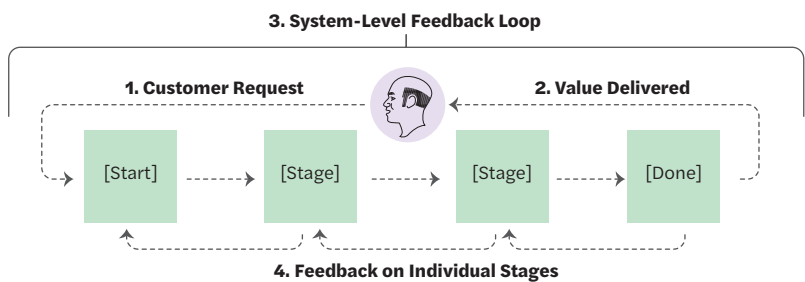


FIGURE 11.1: A Value Stream Map Depicts Multiple Feedback Loops



The cycle from customer request (1) to value delivered (2) establishes a system-level positive feedback loop (3) that encourages growth and allows the team to learn and tune performance. Internal feedback loops (4) are typically negative feedback loops that allow for more fine-grained control and learning.

FIGURE 11.2: Business Frameworks Are Templates for Management

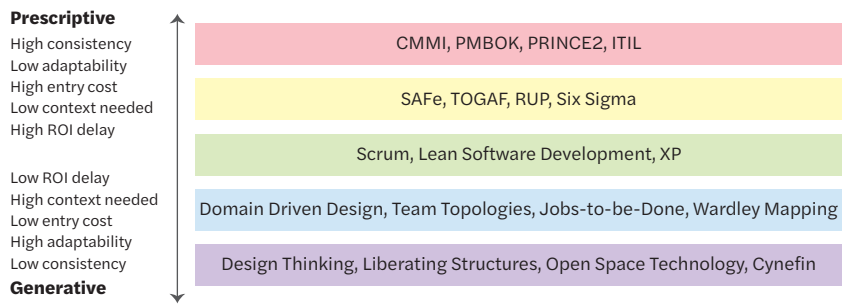
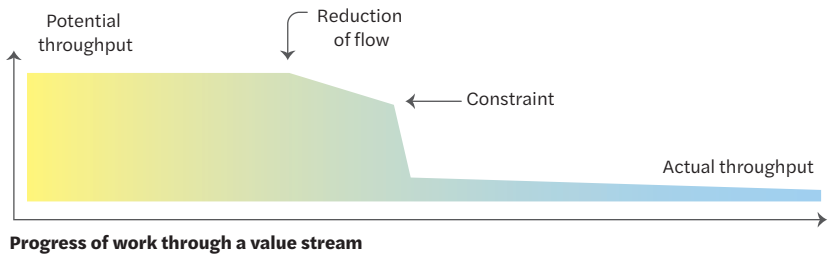


FIGURE 11.3: The Constraint on a Value Stream



This figure illustrates how a constraint has a distinct, measurable impact on the flow of work on a value stream.

TABLE 11.1: Examples of Enabling and Governing Constraints

Enabling	Governing
Maps/canvases	Service level agreement
OKR/V2MOM structure	Industry regulation
Technology platforms	Organizational structure and hierarchy
Playbooks	WIP limit
Principles	Security policies

FIGURE 12.1: Outcome Discovery and Outcome Mapping
Help You Craft a Productive Narrative

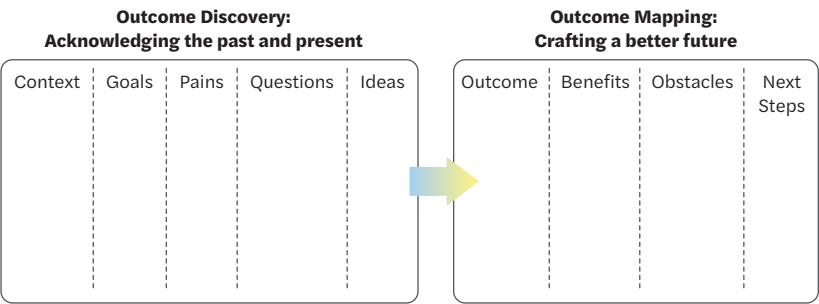


FIGURE 12.2: Outcome Mapping and Discovery Can Bridge Local Incentives by



Establishing a Higher-Value Shared Target

FIGURE 12.3: Limited Visibility Enables Limited Focus

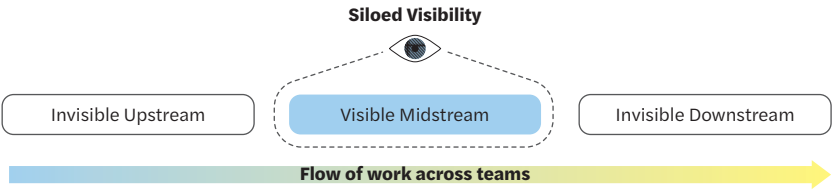
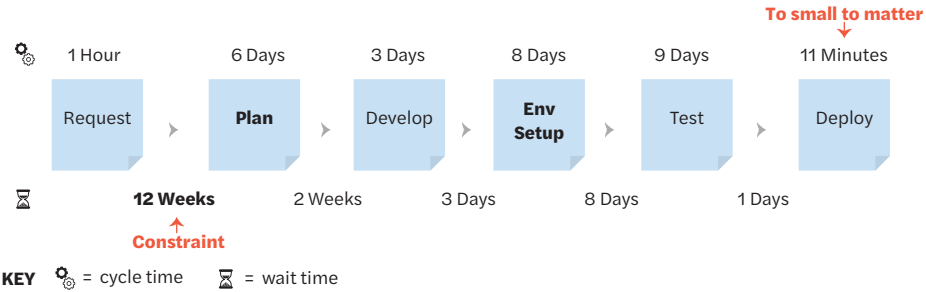


FIGURE 12.4: Example Value Stream Map with Irrelevant Downstream Precision



When your constraint adds weeks of delay, a few hours difference isn't consequential.

FIGURE 12.5: The Current State Map Depicting the Full Agile Release Train (ART)

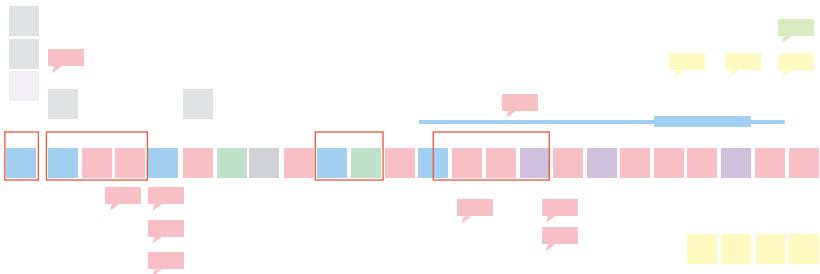
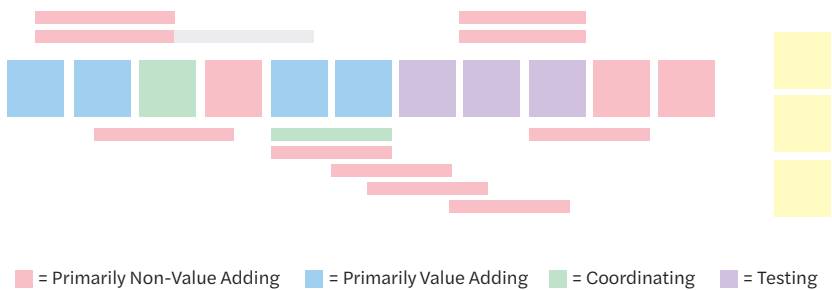


TABLE 12.1: Visual Activity Classification Legend, Depicting Non-Value-Adding, Testing, Value-Adding, and Coordinating Activities

Activity Class	Definition	% of Lead Time	Color
Non-value-adding	Doesn't add stakeholder value to the outcome	58%	Pink
Testing	Validating work (non-value-adding)	14%	Purple
Value-adding	Creates or enhances value from a stakeholder perspective	18%	Blue
Coordinating	Planning, organizing, scheduling, not including 3–6 months of pre-planning (non-value-adding)	10%	Green

FIGURE 12.6: Example Future State Map Depicting Non-Critical and Non-Value-Adding Activities



The Future State Map depicts non-critical and non-value-adding activities executed in parallel rather than blocking flow.

TABLE 12.2: Current State Flow Performance vs. Future State

Current State	Future State
Unclear improvement opportunities	Single constraint to target
Optimized for edge cases	Optimized for common cases
\$2M per release	\$1M per release
64-week lead time	32-week lead time (with no stage modification)
Non-value-adding stages blocking progress	Non-value-adding stages executed in parallel
Value-adding activity: 17% of lead time	Value-adding activity: 53% of lead time

FIGURE 13.1: Focus on Project Leads to Ignoring Longer Term Impact



FIGURE 13.2: Each Cycle of Continuous Improvement Builds on the Previous One

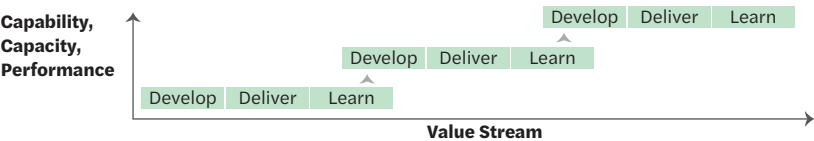


FIGURE 13.3: Current State Map: Steps, Timing, Roles, Tools, Artifacts

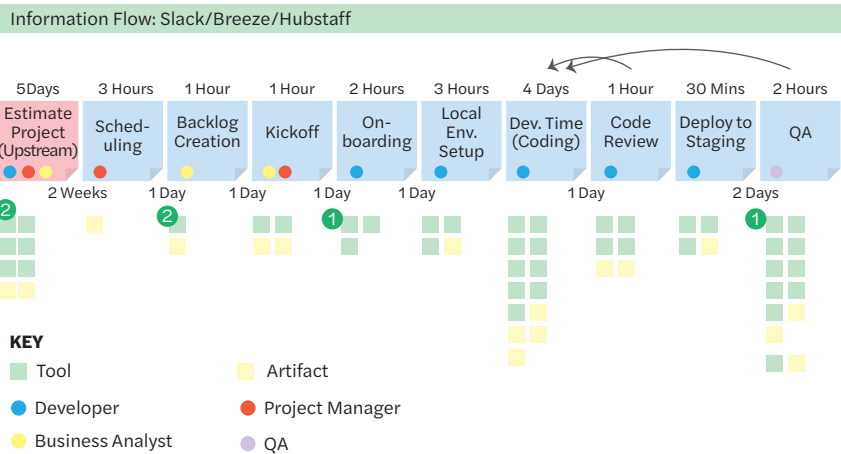
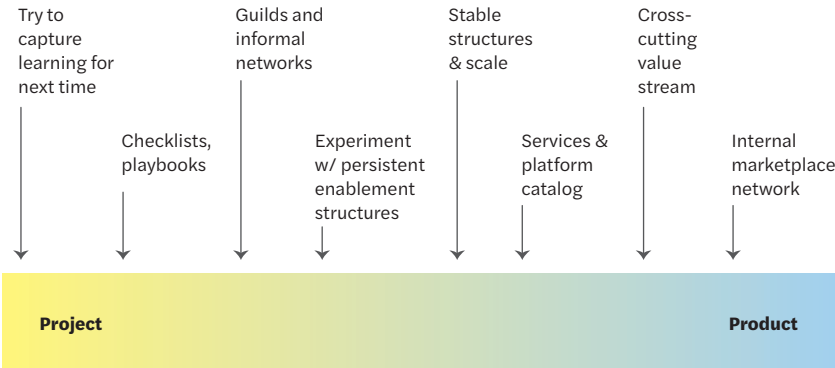


FIGURE 13.4: Project or Product Orientation Lies on a Spectrum



- | | |
|---|--|
| <ul style="list-style-type: none">• Initial value bet• Bring people to the work• Temporary organization• Single batch• Output oriented• Customer validated | <ul style="list-style-type: none">• Adaptive value bets• Bring work to the people• Continuous, iterative• Small batches• Outcome oriented• Customer collaborative |
|---|--|

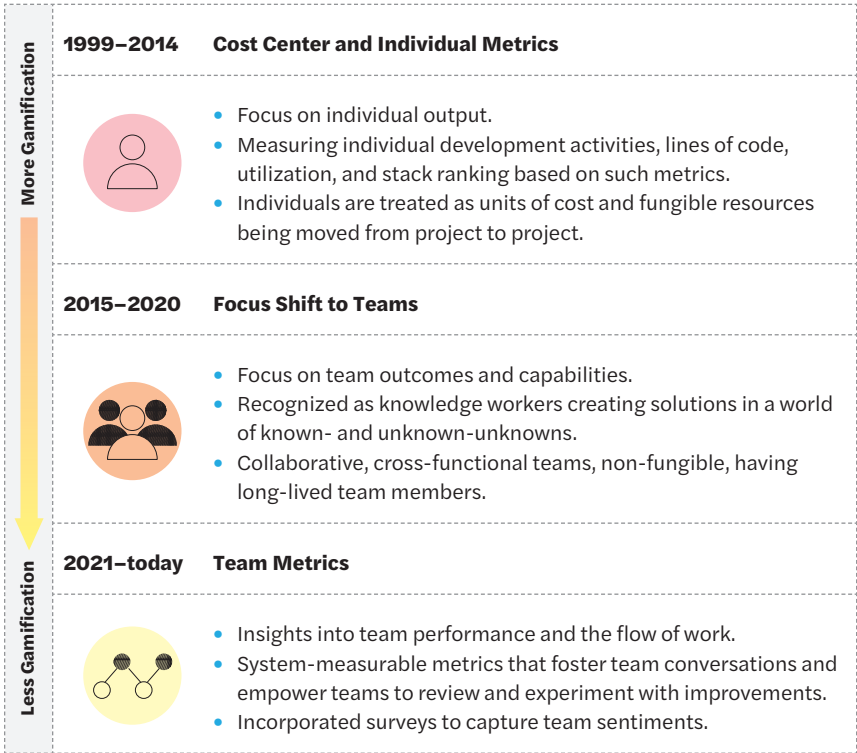
TABLE 13.1: A Comparison of Value Stream Mapping and Value Stream Management

Value Stream Mapping	Value Stream Management
Iterative and periodic practice	Continuous practice
Easy to start	Difficult to start
Human-centric, storytelling, and narrative	Tool- and data-centric, automated insights

TABLE 13.1: A Comparison of Value Stream Mapping and Value Stream Management *cont.*

Value Stream Mapping	Value Stream Management
Perspective and qualitative focus	Perspective and qualitative focus
Visual collaboration	Visibility and continuous inspection
Walls and sticky notes (also virtual)	Integrations, dashboards, and data
Creates and recreates business case	Ongoing incremental optimization
Highlights waste and assumptions	Automates for continuous compliance
Defines present and future	Records past and present
Facilitates creativity, reinvention	Facilitates improvement, iteration
Future opportunity: living representation	Future opportunity: simulation & prediction
Enabling constraint	Governing constraint

FIGURE 13.5: Phil's Journey at Parchment



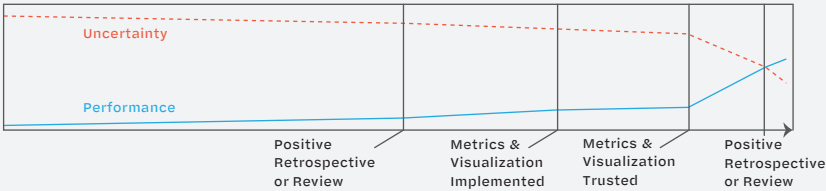
Phil's metrics journey includes eleven years with Parchment. It showcases a twenty-four-year evolution from individual metrics to a combination of team and system metrics, along with team sentiment.

TABLE 13.2: The Parchment Measurement Journey

Individual Scrutiny	Collective Flow
Lines of code	Outcomes over output
Utilization	Team metrics
Comparing individuals based on individual output	Projects to products
	Value Stream Management
	System metrics: Flow, DORA
	Sentiment metrics: SPACE, DX, etc.

FIGURE 13.6: Waiting for Data Availability and Quality Means Uncertainty Remains High for Longer and Outcomes Are Later to Arrive

Team A Performance Improvement Journey



Team B Performance Improvement Journey

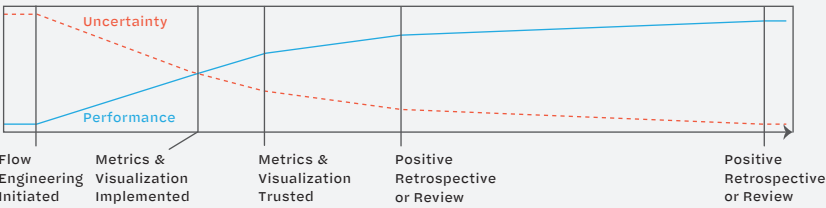
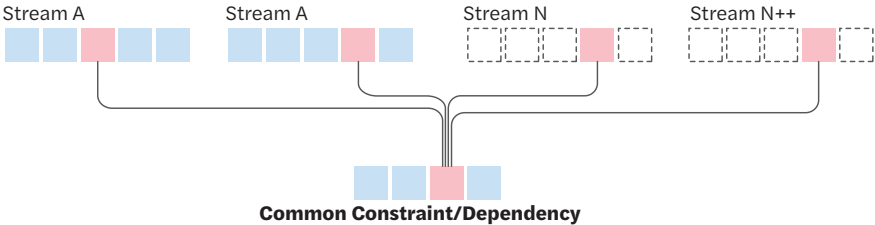


FIGURE 14.1: Value Streams Sharing a Common Constraint



Many value streams share common constraints. Once a common constraint is identified, it can be improved for all dependent streams.

FIGURE 14.2: The Spectrum of Flow Investment

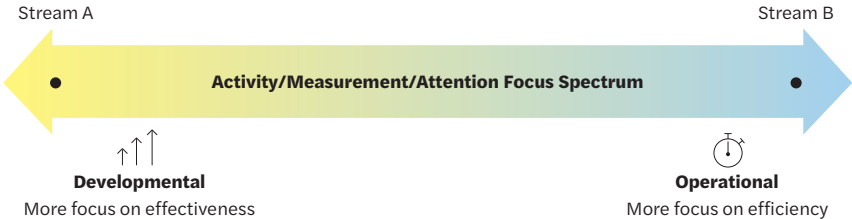


TABLE 14.1: Developmental and Operational Focus Implications

Enabling	Developmental	Operational
Focused on:	Effectiveness	Efficiency
	Design	Execution
	Innovation	Stability
	Novelty	Maintenance
	Value creation	Value maximization
Characterized by:	Variation	Consistency
	Uncertainty	Certainty
	Investigation	Measurement
	Experimentation	Optimization
Function of constraints:	Enablement	Governance

FIGURE 14.3: Interdependence of Core and Supportive Value Streams

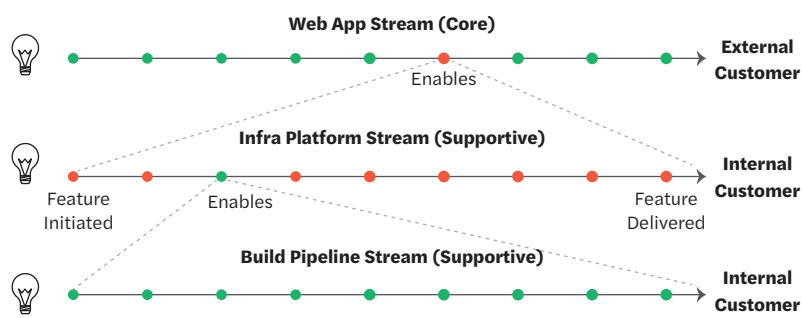
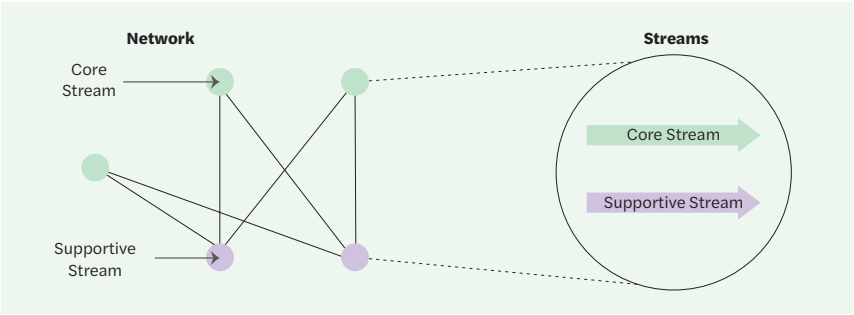
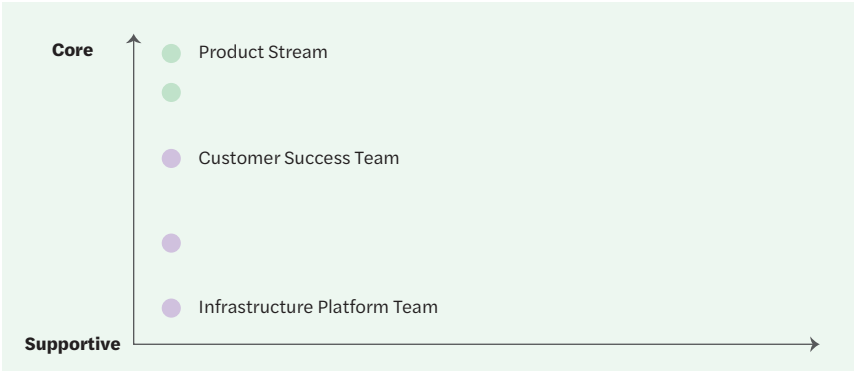


FIGURE 14.4: A Value Stream Network Map



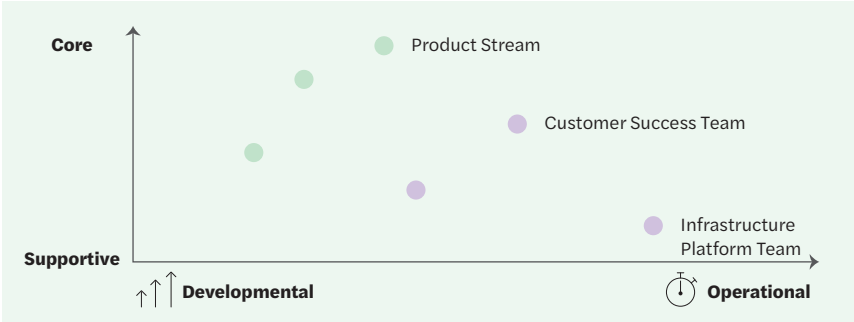
The value stream network can represent interconnected streams at macro scale.

FIGURE 14.5: The Y Axis of a Value Stream Network Map



Supportive to core denotes proximity to external customers.

FIGURE 14.6: The X Axis of the Value Stream Network Map



Developmental to operational denotes the performance focus of the stream.

FIGURE 14.7: Connections Show Dependencies between Value Streams

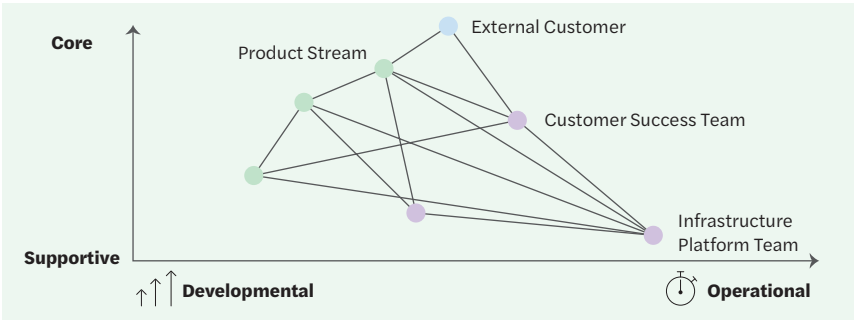
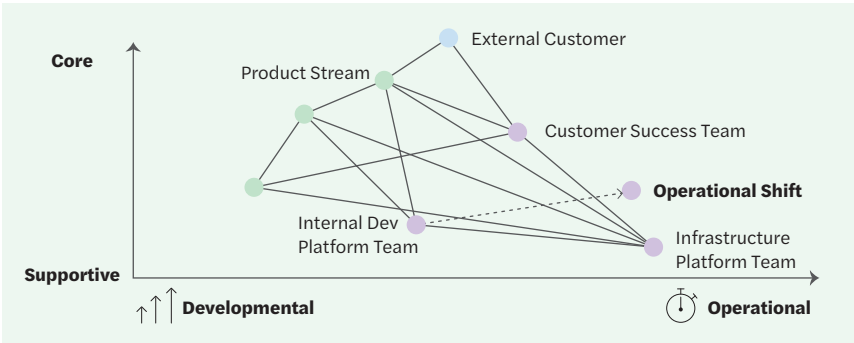
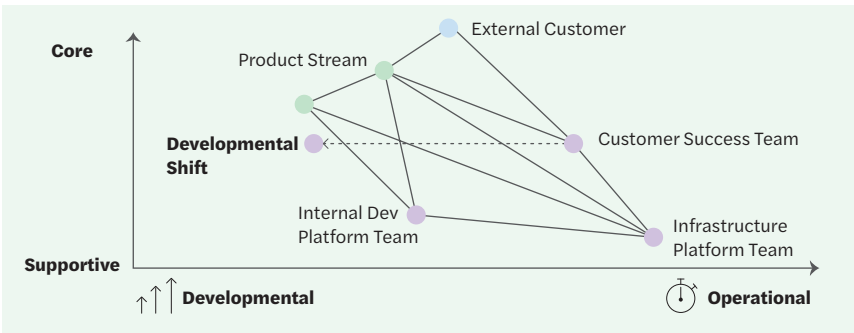


FIGURE 14.8: Example Value Stream Network Map Showing Shift in Internal Dev Platform



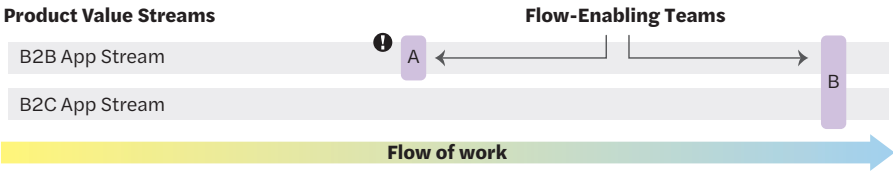
Streams can shift performance focus to respond to needs or conditions, such as an increase in stability or cost reduction.

FIGURE 14.9: Example Value Stream Network Map Showing Developmental Shift



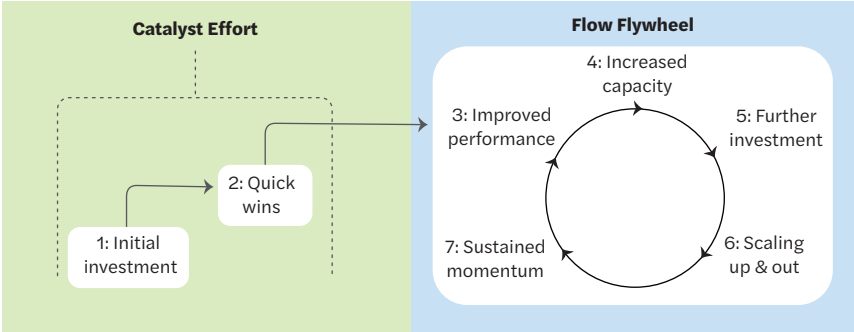
Streams can shift to focus more on innovation or new capabilities, which has implications for how they're measured and managed.

FIGURE 14.10: Flow Enabling Team Interactions with Product Streams



A flow-enabling team works with value stream aligned teams to help them learn, map, and address flow issues. It can support either reactively (A) or proactively (B).

FIGURE 14.11: The Flywheel of Flow Engineering

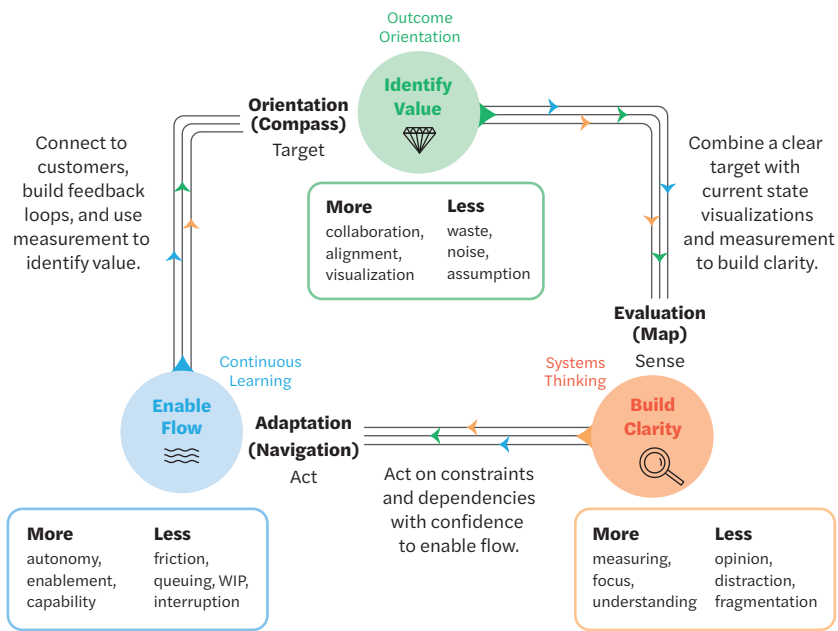


Appendix

Value, Clarity, and Flow Detail

Figure A.1 presents a more detailed view of the three elements of action: value, clarity, and flow.

FIGURE A.1: Value, Clarity, and Flow



Applying Little's Law

Little's Law is a principle from queue theory that provides a simple calculation to evaluate flow in value streams. It helps you understand, for example, how long you can expect to wait for items in your backlog to be delivered based on current performance. Little's Law relates the number of items waiting to be processed (L) to the average arrival/departure rate (λ) and the average wait time of an item within the system (W). The law is expressed as $L = \lambda W$.

In knowledge work, it can be used to relate the amount of work in progress (WIP) to the time to complete this work. It can help you calculate optimal WIP throughout your value streams by showing the combined impact of ingress work, WIP, and throughput. To calculate this correctly, you must ensure you are familiar with the average values for your value stream prior to performing calculations: lead time (W), work in progress (L), and throughput (λ).

As Daniel Vacanti pointed out, being able to use it reliably requires consistent measurement and “looks at the past, not the future.” Little's Law represents a mathematical ideal that depends on a number of assumptions:

- This law holds only for value streams operating at a relatively steady state. As an example, the WIP at the start of the week should match the WIP at the week's end.
- The average rates of both incoming and outgoing work through the value stream should be equal, otherwise throughput can't be accurately quantified, i.e., work items can't be injected or abandoned midstream.

Due to the high amount of variability in knowledge work, these assumptions rarely hold true. But the general relationship between WIP, lead time, and throughput is directionally correct: if you want to increase throughput, work to reduce WIP and/or lead time.

Components of Little's Law

- **L:** Average number (load) of work items (features, bugs, improvements, etc.) in process (WIP) in the value stream.
- **λ :** Throughput of work items through (initiated and completed) the value stream.

- **W:** Average time a task spends in the value stream, from being initiated to being enabled for use. You can think of W here as customer wait from request to fulfillment.

Scenario: Applying Little's Law in a Value Stream Context

- Observation Period: 1 month.
- Work Items Completed: 60 work items.
- Work in Progress: On average, 15 work items are in different stages of the value stream at any given time.

STEPS

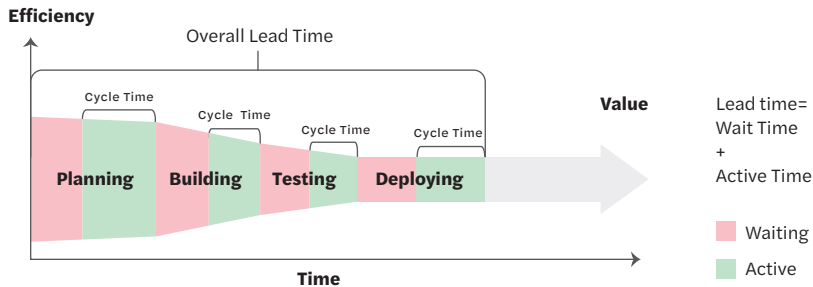
- Calculate the arrival rate (λ): If 60 items are completed in a month, the arrival rate λ is 60 items/month.
- Calculate the average time an item spends in the system (W): Little's Law can be rearranged to $W = L/\lambda$. With (load) $L = 15$ items and (arrival) $\lambda = 60$ items/month. $W = 15/60$ months = 0.25 months.
- This means, on average, a work item takes about 0.25 months (or about 7.5 days) from start to finish in the system.

INSIGHTS AND ACTIONS

- **Backlog Completion:** Knowing throughput can help you think about a backlog in terms of when it could be delivered: If each work item takes 60 days to be delivered (W), and you have five items in progress at a time (L), it will take you 6.5 years to clear out a 200-item backlog.
- **Bottlenecks:** If the team finds the average time (W) too long, they can investigate the constraint in the process. Are items getting stuck in code review? Is there a delay in testing?
- **Capacity:** If W is high, it might indicate a need for more capacity or a change in the development process.
- **Predictability:** Knowing W helps predict delivery times, improving the team's ability to make commitments, but only if flow remains consistent.
- **Process Improvement:** If the team aims to reduce W, they can experiment with different approaches, like limiting work in progress, to see how it affects the average completion time.

Applying Metrics to Value Stream Management

FIGURE A.2: Calculating Speed: Lead, Wait, and Active Times

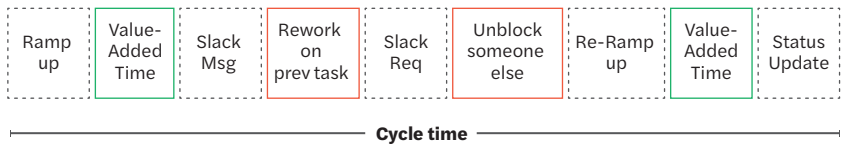


Lead time is the most critical metric in flow improvement for three key reasons seldom discussed:

- Time is the one dimension of work you have the least control over. Lead time is the most complete representation of the fact that time is always passing, for you and your customers.
- Lead time encourages collaboration. It encompasses every cross-functional group involved in value creation, unifying each individual contributor under a collective metric.
- Lead time represents the most complete closed loop of learning from customers. It allows you to understand the earliest point you could deliver value based on an insight gained from your last customer-impacting change.

The biggest misconception of lead time is that it's just the total time that you spend doing the work, which is incorrect. Lead time, by standard definition, actually includes the time that the work waited plus the cycle time for each stage that you worked on that item. So, if you had a user story that was waiting for eight months, and you only spent two months actually developing it, your total lead time ends up being ten months, because we are taking the wait time and the cycle time of each stage together to give you the actual lead time because we are thinking in terms of customer outcomes.

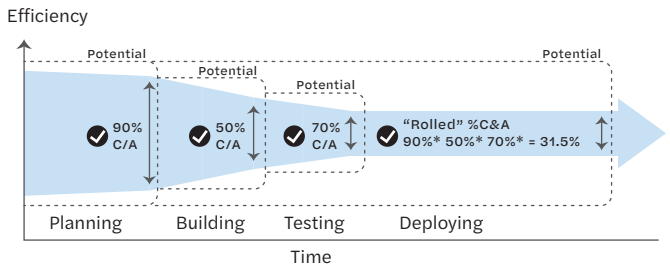
FIGURE A.3: Cycle Time Includes All Value-Added and Non-Value-Added Time Composing Each Stage



Cycle time actually includes the time actively developing the work item along with all the delays, rework, distractions, and operational activities conducted in the course of each stage. A lot of waste can hide in there, but we rarely invest in zooming in to that detail because it’s not necessary or valuable. We simply ask contributors what percentage of the time in each stage is value-added time. Time they’re able to actively develop the value of the work item.

It’s not uncommon for this percentage to be very low, and it should be expected to be low at first. The reason to measure it isn’t to evaluate it at its current level but to establish a baseline for improvement and an understanding of the ROI potential of improvement.

FIGURE A.4: Calculating Quality: Percent Complete & Accurate



% Complete & Accurate: How many items passed through one stage without needing to be returned for rework or clarification? **KEY** ✓ Actual

The image above illustrates how we calculate quality, which is the percent that is complete and accurate. If we break down the definition of this metric, it’s really how complete and accurate the work was that passed through the

various stages, or how many items passed through one stage without needing to be returned for rework or clarification.

At the very beginning of the development process, a product owner or business analyst is typically responsible for gathering requirements and writing user stories. These become work items, which are what the development team then ends up working on. Even an excellent business analyst may only manage to write a user story that hits about 95% of all of the various scenarios and edge cases that need to be addressed. It then gets handed off to a developer. About 5% of the time there are not enough requirements or there's information that is fundamentally missing from the story. At that point, the developer is blocked, and the work actually has to go back to the business analyst.

This happens very often in the DevOps space. The problem that we have is how to measure that and keep track of how often work is being sent back. That scenario can take place not only in the development stage but sometimes even at the testing phase, where someone's completed development, but from a testing perspective, there have been major issues found, so the work gets sent back.

That is where percent complete and accurate (%C&A) comes into the picture and why it is so important. This measure shows what percentage of the work in each of your stages was completed correctly and accurately and did not need to be returned. Ideally, each piece of work would pass from one stage to the next without introducing a defect, but we know that in real-life scenarios and in practice, defects can be introduced at any stage.

The later stages are dependent on the quality of the work in the earlier stages, which means that business analysts really need to be writing clear user stories, including all of the various edge cases. The developer then has to make sure that they are developing the story accurately so that when it gets to testing there are no major issues. As such, we need to make sure that the quality of work is up to par at every single stage.

Let's take one more look at that last image (Figure A.4). You'll see we have our actual percent complete and accurate for planning, which is about 90%; for building, it's at 50%; and then for testing, it's at 70%. So, how do we read that number, which indicates the likelihood that a single piece of work will never have to be reworked in its journey? We can see that our final complete and accurate metric is 31.5%, which is the product of the actual complete and accurate metrics of the planning, building, and testing stages.

With customers who are effectively using Value Stream Maps today, that final number lives somewhere around 15% to 20%, although it's not uncommon to see teams new to Value Stream Management with rolled %C&A under 5%.

Calculating the ROI of Flow Engineering

If you don't currently have clear performance data on your workflow, a clear picture of what it looks like, and a clear path to better, wouldn't you like to have that ASAP? Let's say after mapping you identify three high-priority, relatively simple improvements to implement right away. It will likely take at least three months to see measurable progress. Then, with three more months of progress and operation, you'll have enough data and experience with the current state that you can map it again and look to a new outcome. In six months, you'll have made enough progress that you'll have a new desired outcome, and will have found new bottlenecks to identify and tackle.

FIGURE A.5: Example Map Board in Progress



In practice, Outcome Mapping requires at least one two-hour session to go from scratch to a fully formed map ready for action. We can see in Figure A.5 a very extensive Outcome Map, which covers goals and pains, values and principles, prioritization, and experiment definition. As you go from the raw input toward insight, you can rearrange the maps as you like and easily export them to be shared throughout the organization.

Since knowledge work is increasingly done across time zones rather than shared offices, collaborative online mapping has moved from a bonus to a necessity. With Flow Engineering we're concerned with improving the flow of work in the next three to six months within a specific value stream. Our aim is to not only improve operations but outcomes. Common desired outcomes we've seen in Flow Engineering efforts with clients and partners include:

- Deliver software three times more often.
- Deploy changes to test environments ten times faster.
- Recover from production issues five times faster.
- Reduce production defects by half.
- Reduce variability by half.

One reason that teams (especially remote teams) may hesitate to carve out time for a mapping exercise is that they are afraid to set aside or request sufficient time for the teams to meet. This is a perfect example of false economies, and we need to counter that tendency clearly.

The highest-paid members of any organization are the executive leadership team. But those are also often the teams that most regularly have long team off-sites for thinking and planning and multi-hour meetings each week to plan and strategize. Thinking takes time. Communication takes time. Time costs money. But the cost of inadequate thinking and communication vastly outweighs the up-front cost of thinking correctly from the outset.

The amount of time, money, energy, and goodwill that are burned through ineffective organizational activity is certainly hundreds of billions, if not trillions, of dollars globally. This effect is often described as expending organizational energy as heat instead of as work. Terms like “thrashing” are also used to describe the activity of staying busy without having an optimal impact.

A perfect example of thrashing is people whose calendars are filled with thirty-minute meetings pertaining to a huge number of parallel initiatives. Such meetings leave little time to accomplish real work, and even with buffer time, it takes an average of twenty-three minutes and fifteen seconds to effectively “get back to work.”²² Parallel projects invariably drag on for far longer and yield far lower quality than selectively taking up projects and completing them as quickly as possible before taking on new work.

We can estimate direct and indirect costs of a member in an IT organization being brought into a meeting at \$125 an hour (opportunity cost could be higher). For the mapping exercises described in this book, it's reasonable that they might involve eight people each. Some exercises, like Dependency Mapping, may not be required depending on what's discovered in the earlier stages. This allows us to estimate a cost of \$1,000 per hour of Flow Engineering exercises.

To estimate the benefits, we can imagine that we are influencing the activities of twenty knowledge workers who are each doing work valued at \$200,000 per year (value delivered should be greater than salaries paid). Our goals in Flow Engineering are to help the team increase their throughput, improve quality, reduce attrition by increasing job satisfaction, and reduce the chances of a project being unsuccessful. Assuming they are all working full time, that team should be bringing \$4 million of value per year.

TABLE A.1: Estimated Costs of Flow Engineering Exercises

Activity	Typical Initial Duration	Subsequent Time Required	Estimated Annual Cost
Outcome Mapping	3 hours	1 hour	\$4,000
Current State Mapping	3 hours	1 hour	\$4,000
Dependency Mapping	2 hours	1 hour	\$3,000
Future State Mapping	2 hours	1 hour	\$3,000
Flow Roadmap	3 hours	1 hour	\$4,000
Total	13 hours	5 hours	\$18,000

If we can help the team to increase their throughput by 20% (the common average improvement impact delivered by mapping³), that \$18,000 investment brings an equivalent of \$800,000 benefit per year. Similarly, if we can prevent 20% of their work from being wasted or ineffective, that's worth \$800,000 per year. Reducing attrition also brings financial benefits by averting the need to rehire, retrain, and burden the team with disruption.

That back-of-the-envelope estimate shows a 44x benefit from these activities. This cost accounting approach is obviously an oversimplification, but it can be a useful way to illustrate the cost benefit. What's also notable is that improvement in one stream is typically replicable across multiple streams, magnifying the impact for far less investment. Even if the costs were scaled by several times or the benefits reduced significantly, this represents a remarkably positive effect from investing this time.

Glossary

The most critical aspect of language is that everyone involved uses and understands it the same. We encourage you to discuss what critical terms mean in your context and adjust them as needed. These are provided from our context and can serve as a baseline if you need a reference.

Customer: An entity that receives and perceives value from the value stream's outputs. This encompasses a broad range of recipients, including external clients, internal stakeholders, and partners, highlighting the diverse perspectives of value reception.

Cycle Time: The duration a stage requires for completion, including any inherent delays or waiting periods.

Flow Efficiency: The proportion of active time (or touch time) compared to overall lead time within the value stream, offering a gauge for assessing the stream's operational efficiency. This metric should be placed in the context of blocks, interruptions, rework, and other detrimental factors affecting performance. It can be refined to emphasize the contribution of direct value-generating activities to the stream's performance.

Lead Time (or Flow Time): The comprehensive duration required for a work item to traverse the value stream from start to finish. This metric encapsulates all stages, including production, pauses, and corrections, summarized simply as "hypothesis-to-happiness," signifying the journey from initiation to customer satisfaction.

Percent Complete and Accurate (%C&A): A quality metric indicating the proportion of work that progresses through a stage that meets requirements, without the need for rework or clarification.

Stage: A single distinct activity within the value stream, typically bounded by handoffs. For our purposes, analogous to step, process, or set of tasks.

Segment: A subset of the value stream consisting of multiple stages.

Throughput (or Flow Velocity): The measure of a value stream's output over a given period, indicating the efficiency of delivery to customers. The Theory of Constraints places this in relation to costs. Little's Law places this in the context of input relative to output.

Value: Ultimately defined by the recipient. The benefit that a product or service provides to the customer.

Value-Added Time: The duration spent on activities that directly contribute to the final product or service's value from the customer's perspective.

Value Stream: An integrated series of activities designed to deliver a specific outcome for a *customer* who might be the final consumer, stakeholder, or user. This stream encapsulates the journey of work from initial need to the delivery of value.

Wait Time: The interval of inactivity between stages, characterized by delays and queuing.

Waste: Activities within the value stream that consume resources without enhancing the product or service's value.

Work in Progress (WIP, Flow Load, or Work in Process): The quantity of work items currently under process within the value stream, from the inception of effort to completion. This is often tracked per stage to highlight accumulation and reveal a constraint.

Work Item Profile (or Flow Distribution): The classification of work items within the value stream, aiding in the assessment of work type distribution and flow characteristics. Common profiles include features, experiments, bug fixes and issue remediation, flow improvement, technical debt payment, and risk mitigation.

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